

The CPU: A very tall pile of simple

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You can hear the phrase

"computers think in 1s and 0s"

a hundred times and still not understand how a computer actually works. By itself, this explains basically nothing. Sure, a wire can be high or low. Sure, a light can be on or off. But how does that become addition?

How does that become memory?

How does that become a program sitting in RAM, one instruction after another, telling a machine what to do?

This article is going to walk you through how a CPU is built, starting with the simplest possible components.

We start with a simple circuit turning a light bulb on and off, then work our way through logic gates, memory, and the basic circuits underneath them.

Some resources stay extremely high-level, so you never really understand how a CPU actually works.

The deeper resources are amazing, but they are long, dense, and intimidating. And frankly, for someone who doesn't want that level of detail, a lot of it can often feel like too much. This article will hopefully help you understand what is going on under the hood, without exploding your brain or eating weeks of time.

The key point is that nothing here is smart in isolation. A CPU is not one hard idea. It is a very tall pile of simple ones.

(Full simple CPU drawing: a few labeled boxes, data bus, address bus, and some control wires)

We are going to try to understand this simple CPU. It is not a modern CPU with decades of optimization, but it has the same core functionality.

1 How to Read This Article

- If a diagram is hard to understand, click it and step through each frame one by one using the arrow keys, with the provided descriptions. This only works if you are reading on my website.
- Don't try to "memorize" every layout. Focus on the mental models, concepts, and what part each piece has to play.
- If a section feels dense or hard to understand, follow the diagrams first and try to get a feel for what is happening.

2 Circuits & Electricity

First, we need the basics of how electricity and circuits work for the purposes of this article.

Here is a simple circuit:

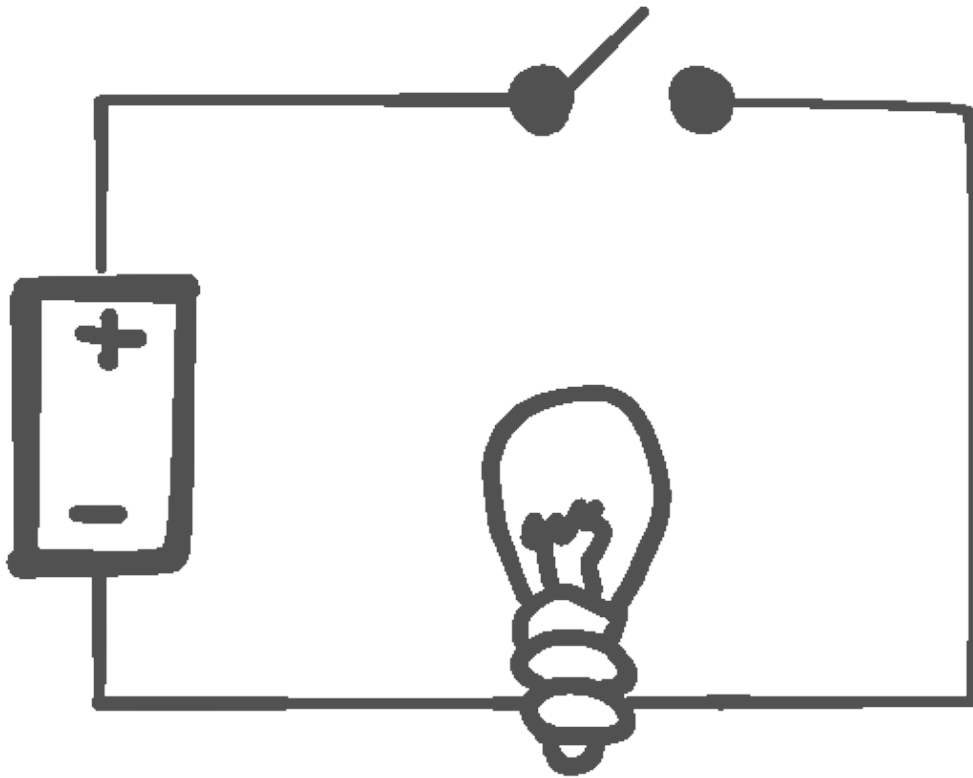


Diagram 2.1. A basic circuit with a battery, switch, and bulb.

We can think of the battery as being able to push charge around the loop. Current can only flow when this loop is completed.

If the loop is broken, nothing flows. A switch is simply a controlled break in the loop, allowing us to break and complete the loop whenever we want.

And a light bulb is just a simple light bulb. It glows when current flows through the filament.

Now we have a circuit that can do one yes/no thing. Current flows or it doesn't.

Now let's see if we can combine switches and relays so the circuit can "answer" slightly more interesting questions.

3 Switches, Relays, & Logic Gates

Say we want to build a simple dog washer circuit: a circuit that, based on some inputs, can tell us whether to wash our dog or not.

Our simple circuit is going to use a light bulb being on to mean yes, wash the dog. Light bulb off means no, don't wash the dog.

So we start with an extremely simple version with two switches.

In this first version, the switches are directly inside the bulb circuit. The person using the circuit can open or close each switch to answer a yes/no question.

Let's say switch 1 represents STINKY: whether the dog is stinky or not. Switch 2 represents OLD_WASH: has it been more than 5 days since the last wash.

So the rules for our first circuit are:

if STINKY AND OLD_WASH, the bulb is on.

Or in other words, if the dog is stinky and its last wash was over 5 days ago, then wash the dog.

Here is the circuit:

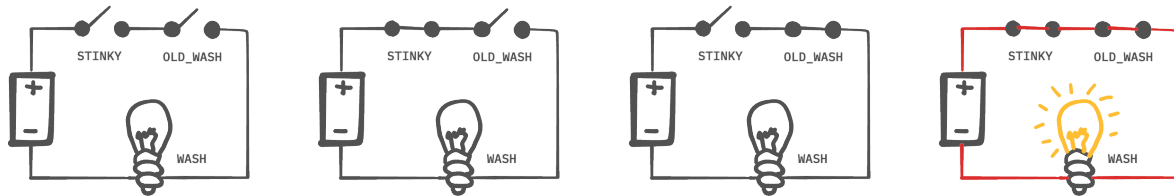


Diagram 3.1. The hand-switch version of AND.

This circuit shows a logical AND operation. A person is flipping the switches manually. The output turns on only when both inputs are true.

Now add a new input: MUDDY, if the dog is muddy.

Now the rules of the circuit change:

if (MUDDY OR STINKY) AND OLD_WASH

All this says is, if the dog is muddy or stinky and it's been at least 5 days since the dog's last wash, you should wash the dog.

Now let's focus on the (MUDDY OR STINKY) part of this circuit:

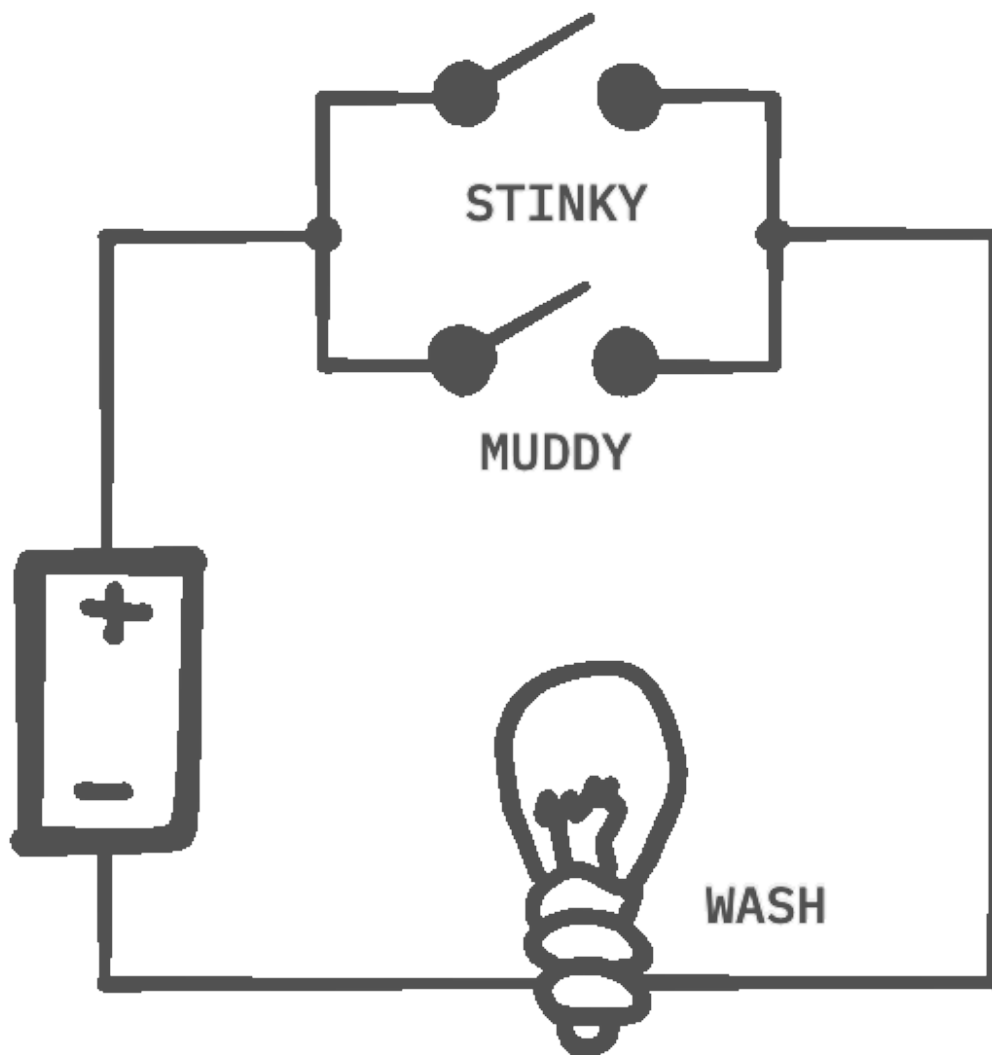


Diagram 3.2. The hand-switch version of OR.

This is a logical OR: either MUDDY or STINKY needs to be on for the bulb to turn on.

Now let's combine the two to form the complete circuit.

But now we have a problem.

The MUDDY OR STINKY circuit outputs its result with an electrical signal: on or off. Our previous AND circuit relies on a human flipping a switch in order to compute a result.

Or in other words, the OR circuit we built outputs a result as electricity, but the AND circuit we want to combine it with expects an input as a metal switch physically being moved. A signal in a wire can't reach over and close that switch by itself.

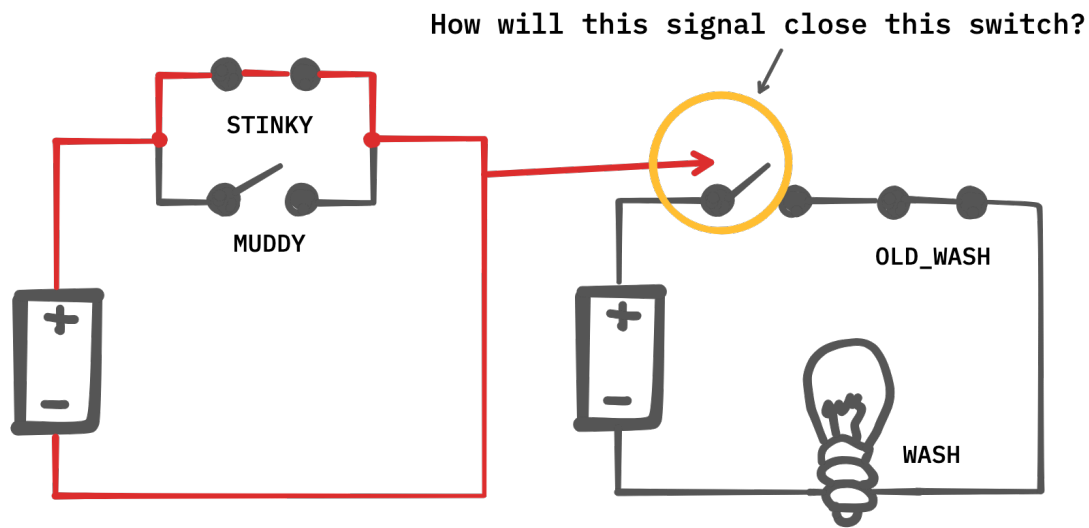


Diagram 3.3. An electrical signal cannot move a metal switch by itself.

So if we want to chain circuits together, we need a way for an electrical signal to control a switch automatically. How can we do this?

Electromagnetic relays, that's how. Or at least, that is one early solution to this problem. We will talk about other solutions a little more later on.

This probably sounds quite complicated, but it is just a magnet powered by electricity.

Here is how it works:

One thing to mention: if you see several little batteries in a circuit, don't interpret that as several totally separate power sources. I am using the battery drawing as a symbol for "this point is connected to power," so the diagram doesn't turn into spaghetti.

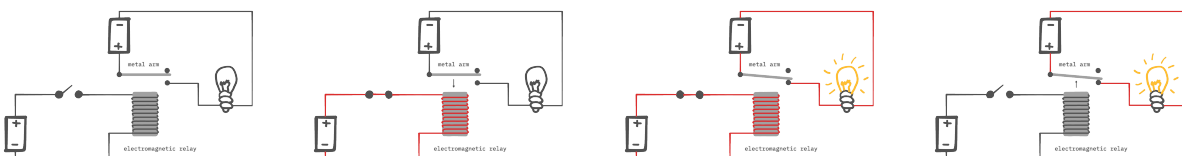


Diagram 3.4. An electromagnetic relay.

This relay is made from a coil of wire and a movable metal arm. When current flows through the coil, the coil becomes a magnet and pulls the arm down. When current stops, a spring pulls the arm back up.

A relay lets one circuit open or close a switch in another circuit. The two circuits stay separate, but the relay arm physically connects them.

Also, in this example, we end up using a switch in the input circuit anyway, but any kind of electrical signal could be used, like the output of another circuit. The switch is just there to demonstrate how the relay works.

As you can also tell by the diagram, there is a slight delay between the coil turning on and the metal arm moving. Relays are mechanical, so they do not switch instantly.

Now let's see how we can build an actual electrical AND gate that takes two input wires and outputs an electrical signal.

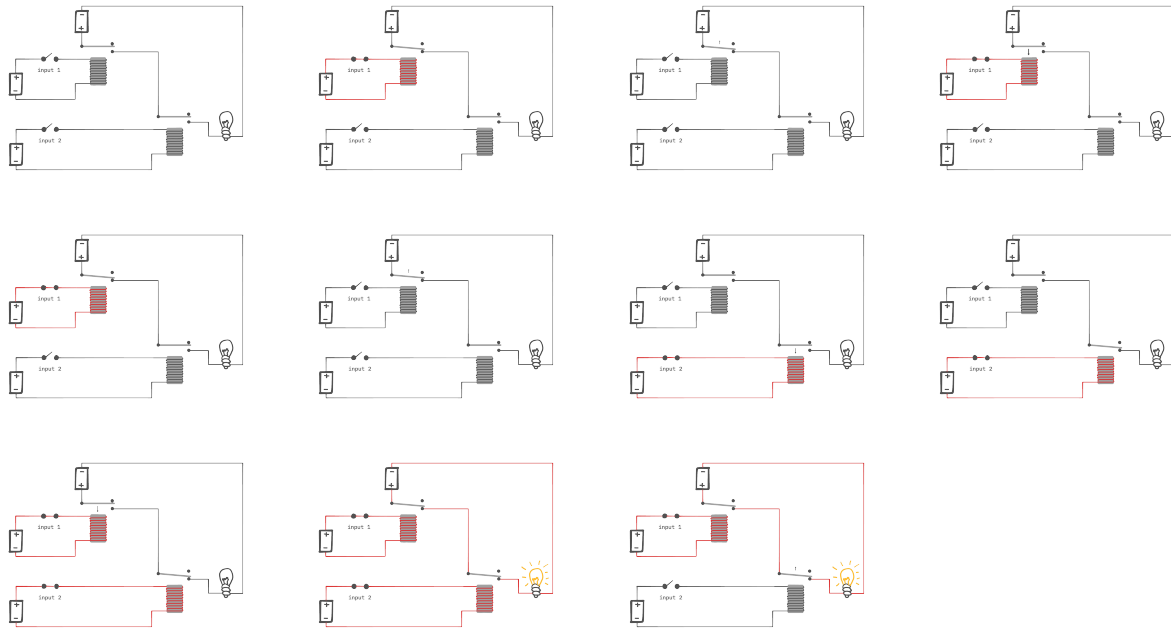


Diagram 3.5. An AND gate.

The output circuit has two breaks in it, one controlled by each input relay. Only when both inputs have signal do both relays close, completing the output loop.

Using these relays chained in clever ways, you can create every fundamental logic gate, such as the OR gate:

But before the next diagram, I am going to use one more new symbol: ground.

For the purposes of this article, the ground symbol will simply refer to the common return point of the circuit, usually connected to the negative side of the battery.

Every point marked with the ground symbol is connected together, as if there were hidden wires joining them underneath the drawing. It is not a new component. It is just a less messy way to draw the return path of the circuit.

The circuits are still loops. I am just not explicitly drawing the return wire anymore.

In a real schematic, the ground symbol itself would usually stay white. In these diagrams, I sometimes color it red when that return point is part of the active path for that frame. I think it makes the current path easier to follow visually.

This is how the ground symbol looks:

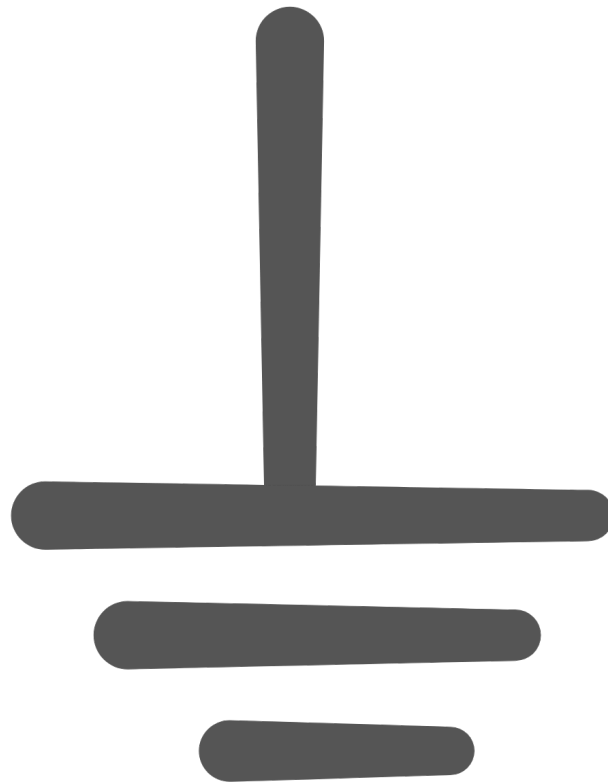
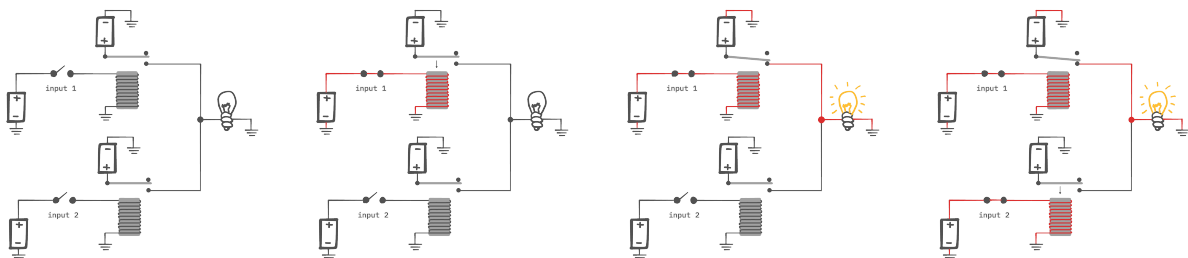


Diagram 3.6. The ground symbol.

Now here is the OR gate:



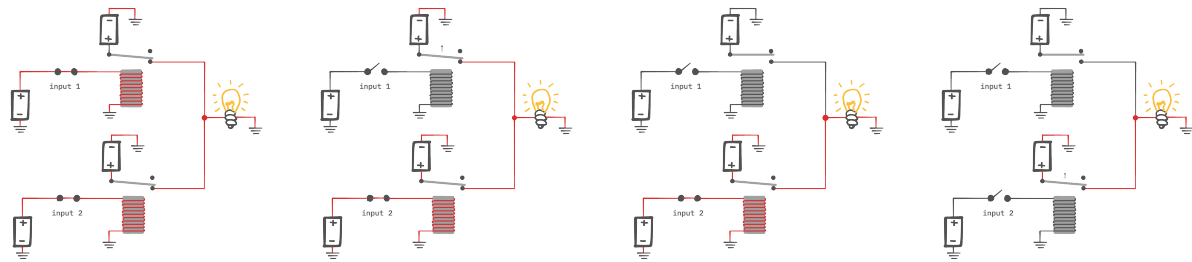


Diagram 3.7. An electronic OR gate.

That is an OR gate using relays. Now here is the full dog washer circuit up to this point:

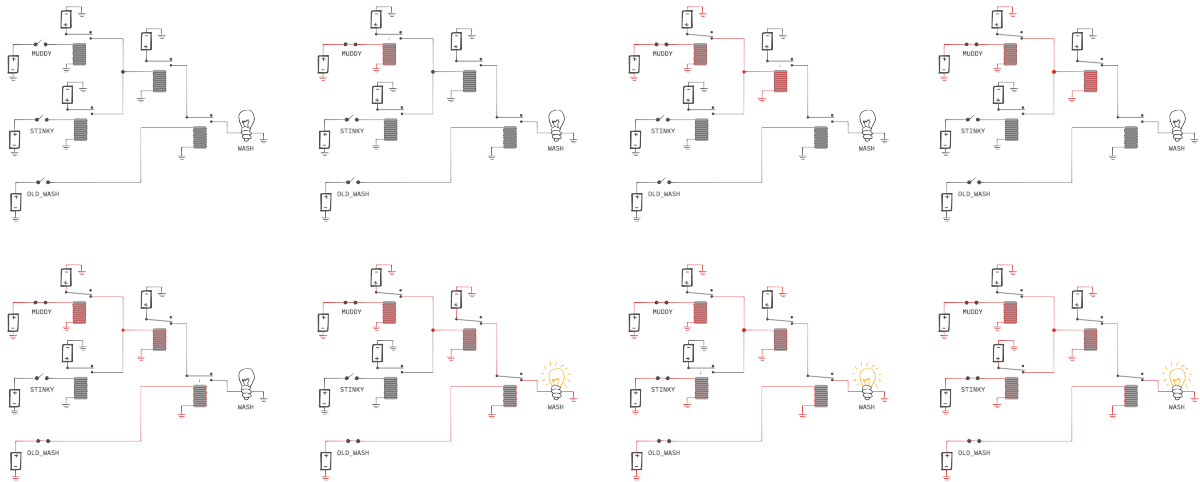


Diagram 3.8. The full dog washer circuit built with relays.

The animation does not show every possible combination of switches, only a handful. But in a nutshell, if MUDDY or STINKY is on, and OLD_WASH is also on, the bulb turns on.

Okay, now let's introduce one last input, or "sensor": RAIN_SOON, whether it is predicted to rain soon. The rules of the circuit change once again:

$((\text{MUDDY OR STINKY}) \text{ AND OLD_WASH}) \text{ AND NOT RAIN_SOON}$

The parentheses indicate order of operations. This should be pretty familiar. So in plain English:

If the dog is muddy or stinky and it's been at least 5 days since the dog's last wash and it's not going to rain soon, then wash the dog.

Let's focus on this NOT for a second. NOT just inverts a signal: if it receives signal, it outputs no signal; if it receives no signal, it outputs signal.

That is what a NOT gate does.

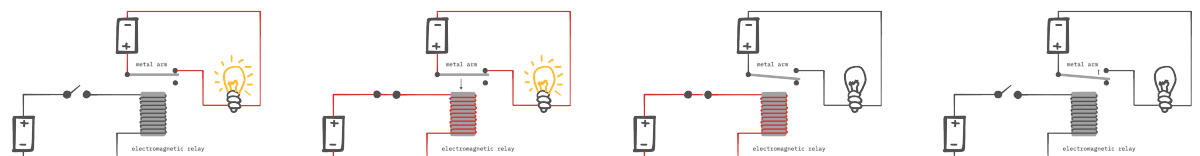


Diagram 3.9. A NOT gate.

Now let's clean up some of our understanding of circuits before we move on. We have been showing our outputs as a light bulb. For a bulb to be on, it needs to be connected to + and -, one on each side. That difference in voltage allows current to flow, turning on the bulb.

But let's say we just want an output wire, not a bulb. We can't just remove the bulb; + connected directly to - would lead to a short-circuit. So what we do is either drive the wire up or down, so it is connected to either + or -. All of our relay gates can be simply adapted to do this.

This distinction matters later. A 1 output is a wire being driven high. A 0 output is not "nothing"; it is a wire being driven low. It will make sense why I am mentioning this early, later.

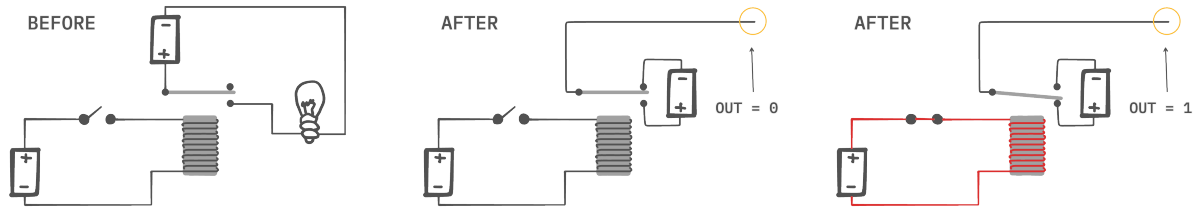


Diagram 3.10. Driving an output wire.

In this diagram, red wire means current is actively flowing, that's why $OUT = 1$ is still white. Later when we stop drawing every logic gate, red wire will just mean high, or 1.

Now before we look at the completed circuit, let's learn some basic logic gate symbols.

An AND gate is drawn like this:

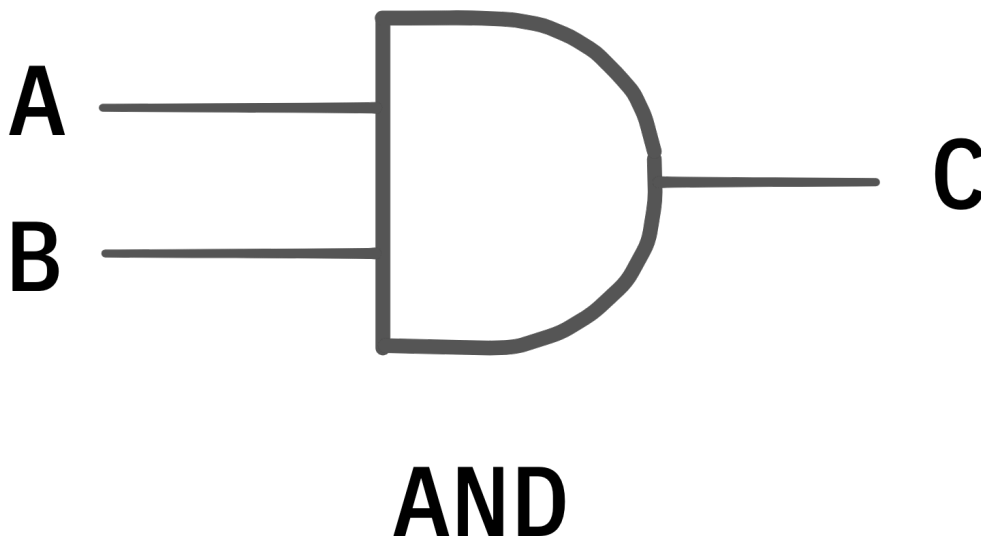


Diagram 3.11. An AND gate.

This symbol represents the AND circuit we made previously, except instead of turning a bulb on and off, it drives an output wire.

An OR gate is drawn like this:

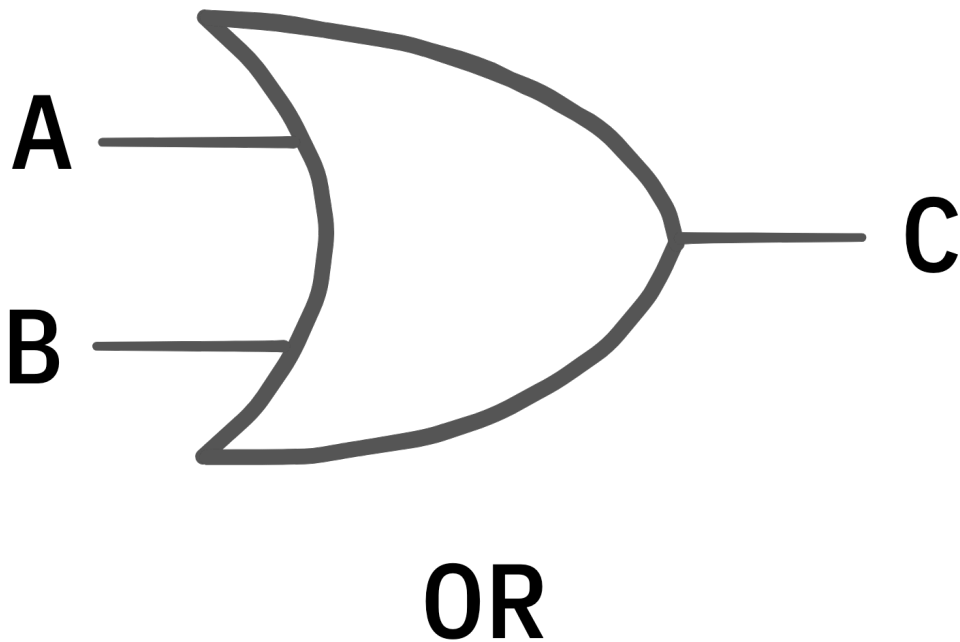


Diagram 3.12. An OR gate.

This symbol represents the OR circuit we made previously, with the same idea about driving the output wire instead of a bulb.

Whenever I use these symbols moving forward, they can almost directly translate to the circuits with the relays I showed you previously, but the internal components stay hidden for cleanliness.

Here are three more useful gate symbols:

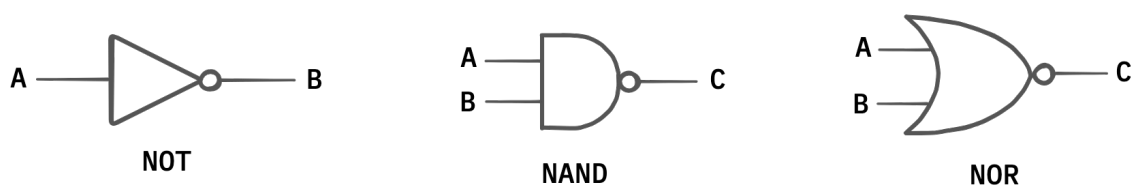
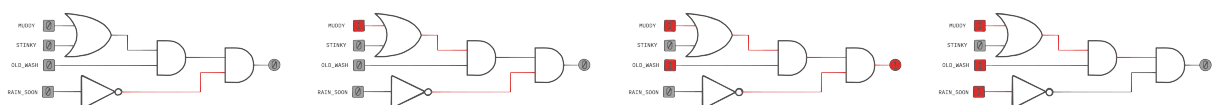


Diagram 3.13. NOT, NAND, NOR gates.

NAND is AND with the output flipped. NOR is OR with the output flipped.

That little circle at the end of a gate means "flip the output."

With our knowledge about logic gates, let's create the "should-I-wash-my-dog 5000" machine!



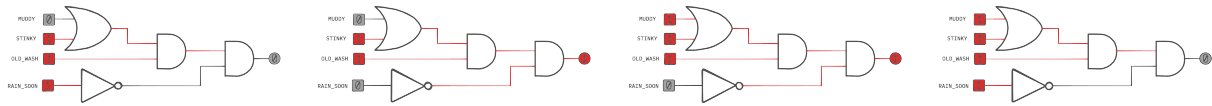


Diagram 3.14. The final dog washer circuit.

Again this animation doesn't cover all possible states.

Keep in mind these electromagnetic relays we used in the examples are quite big and slow.

Relays aren't the only solution. They are simply one of the early and intuitive methods to understand, and many real computers like the Harvard Mark I actually used these types of relays.

In modern computers, similar behavior is achieved by using transistors. If you want to learn more about transistor based logic gates: visit [this site](#).

I don't know about you, but addition seems like a pretty logical next step to these logic gates. But not so fast.

This is how circuits make yes/no decisions. Not by understanding what MUDDY means, but by wiring simple gates so the output turns on only for the input pattern we care about.

A wire is just a wire. We gave these wires meaning. We decided that one wire means STINKY, another wire means MUDDY, and another means RAIN_SOON.

To make a CPU, we need to give wires a different kind of meaning: numbers. Before we can build a circuit that adds, we need a way to represent numbers using only on and off.

That is what the next section is about.

4 Counting With Wires

Okay, before we continue with this section, let's define some terms.

A wire with no signal is 0, and a wire with signal is 1. Let's call one wire, one bit. A bit can either be 0 or 1.

These are just labels that represent the state of a wire.

A group of 8 bits is called a byte. With 8 bits, there are 2^8 , or 256, possible patterns. So if we use those patterns to represent non-negative numbers, one byte can represent 0 through 255.

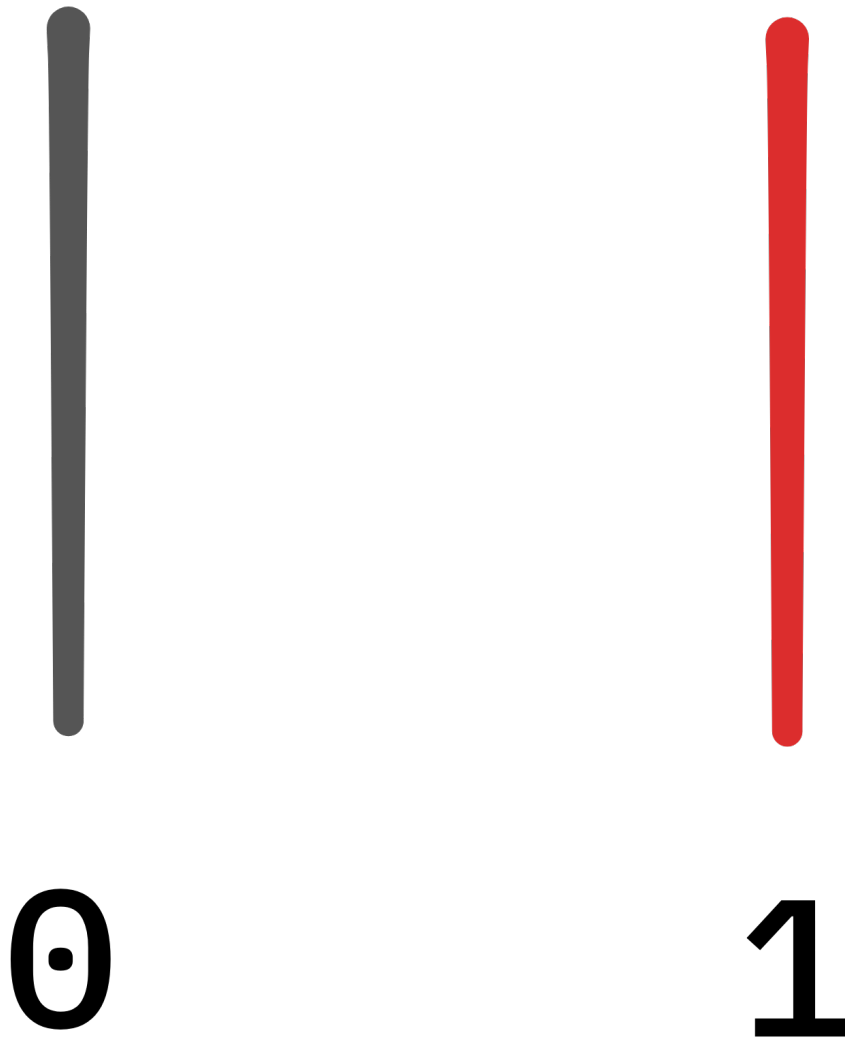


Diagram 4.1. One wire can represent two states: 0 or 1.

If we want to represent numbers using wires, we are going to need more than one wire, because one wire can only represent up to two numbers, since it only has two possible states: 0 or 1.

But two wires have 2^2 , or four states, and three wires have 2^3 , or eight states. That would allow us to represent more numbers.

Here are all the possible states we have with 3 wires:

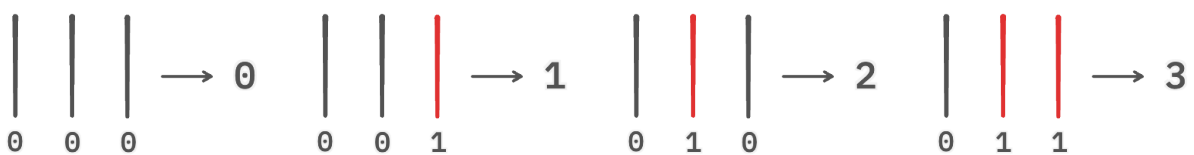




Diagram 4.2. States with 3 wires.

We can represent 8 numbers just like this. The more wires we add, the more numbers we can represent.

But, why does 010 mean 2? Why does 101 mean 5? Is it just randomly assigned?

Not exactly. To understand this, let's take a quick detour to decimal, a.k.a. base ten.

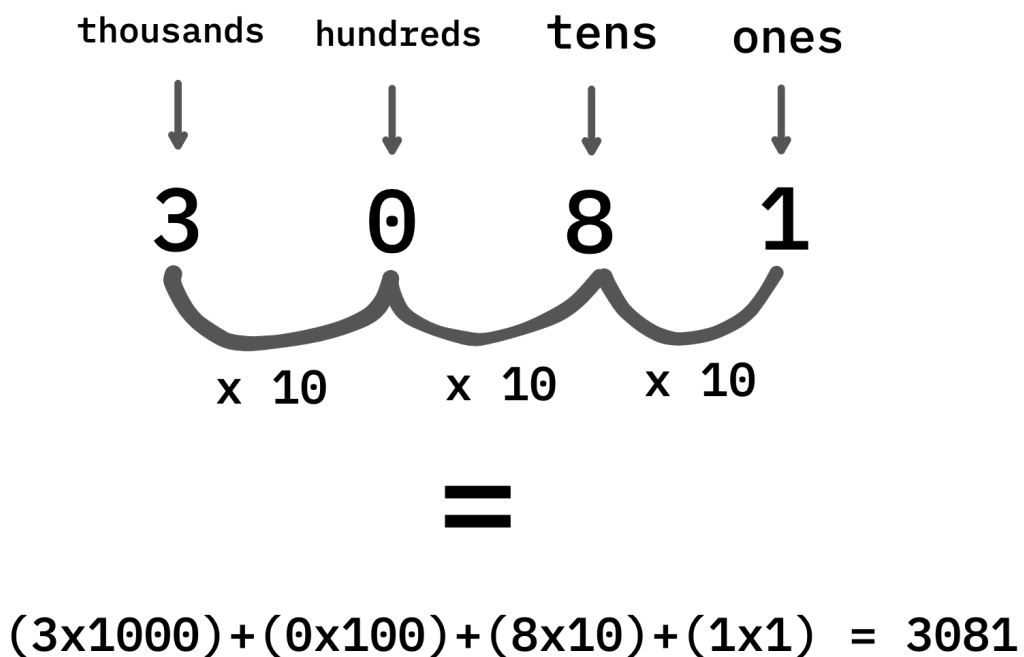


Diagram 4.3. The decimal system.

In our decimal counting system, each place value is a multiple of 10. That is because we have ten digits: 0-9.

This exact same place value logic can apply to the binary system too. We have two digits, 0 and 1, so each place is a multiple of 2.

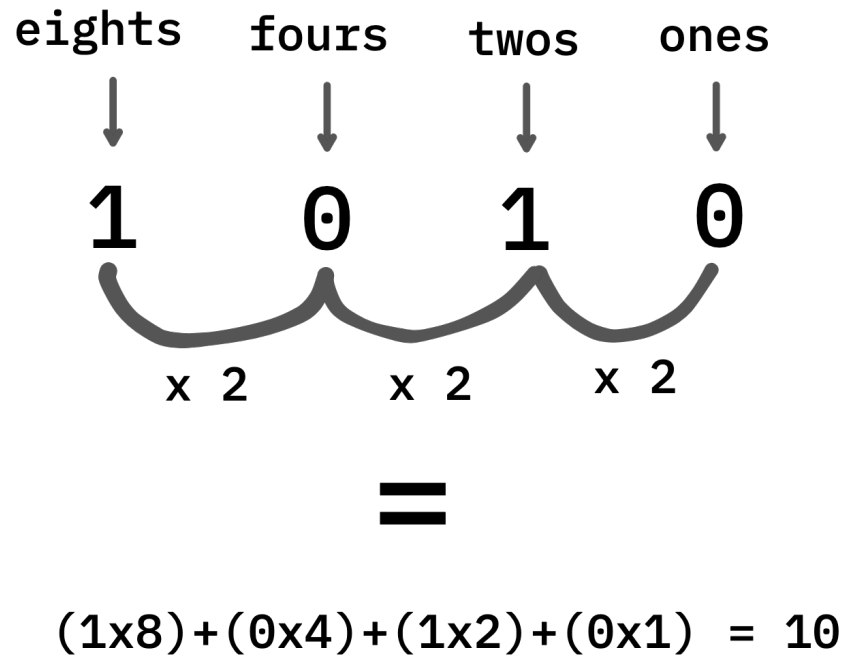


Diagram 4.4. The binary system.

So binary is, at the end of the day, decimal but with only two digits instead of ten.

A few examples:

- 101 means 5
- 1101 means 13
- 101010 means 42
- 1100011 means 99

You don't need to do these problems in your head, but I hope the idea of how binary works makes sense.

Let's walk through 1101 together.

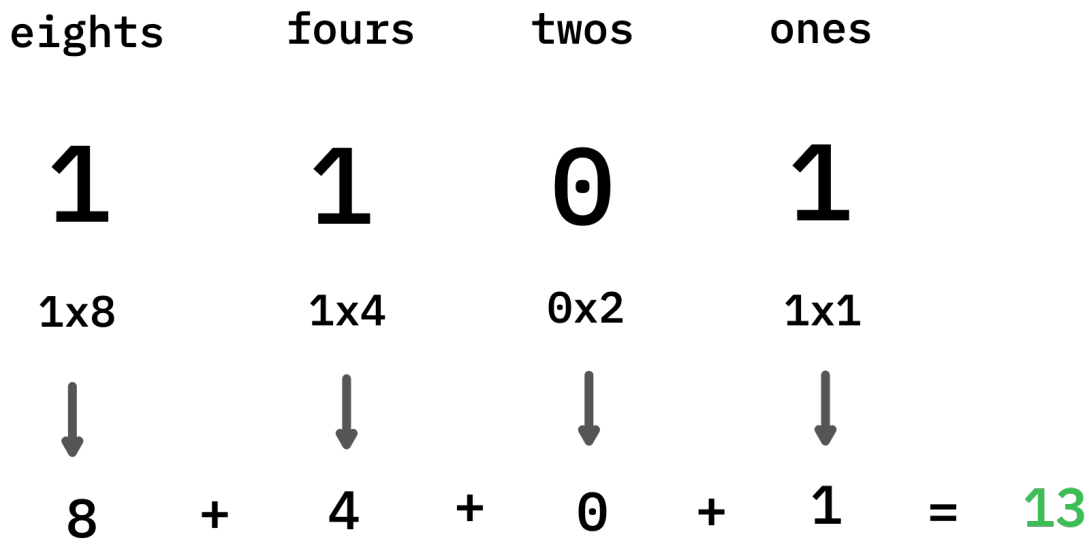


Diagram 4.5. An example in binary.

The binary system works the same way as decimal. The only difference is that instead of multiplying the digit by a power of 10, we multiply it by a power of 2. That's it.

So now that we can represent numbers with wires, how can we add numbers together? How can we compute sums? That is what the next section is all about.

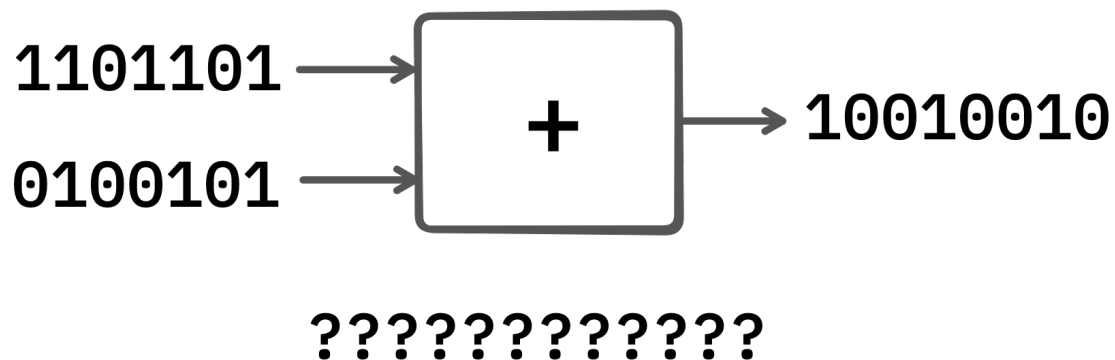


Diagram 4.6. Addition?

5 Addition

Let's start with a brief reminder of how we algorithmically add two decimal numbers.

$$\begin{array}{r}
 65 \\
 + 28 \\
 \hline
 \end{array}
 \qquad
 \begin{array}{r}
 65 \\
 + 28 \\
 \hline
 3
 \end{array}
 \qquad
 \begin{array}{r}
 65 \\
 + 28 \\
 \hline
 93
 \end{array}$$

Diagram 5.1. Standard decimal addition.

We start at the rightmost column, do $5+8$, get 13, we carry the 1. So we write 3 as the sum, and 1 as the carry. We then move left and repeat over and over remembering to add any carry-in values. Binary addition works the same way.

$$\begin{array}{r}
 0111 \rightarrow 7 \\
 + 0011 \rightarrow 3 \\
 \hline
 \end{array}
 \qquad
 \begin{array}{r}
 0111 \rightarrow 7 \\
 + 0011 \rightarrow 3 \\
 \hline
 0
 \end{array}
 \qquad
 \begin{array}{r}
 0111 \rightarrow 7 \\
 + 0011 \rightarrow 3 \\
 \hline
 10
 \end{array}
 \qquad
 \begin{array}{r}
 0111 \rightarrow 7 \\
 + 0011 \rightarrow 3 \\
 \hline
 010
 \end{array}$$

$$\begin{array}{r}
 0111 \rightarrow 7 \\
 + 0011 \rightarrow 3 \\
 \hline
 1010
 \end{array}
 \qquad
 \begin{array}{r}
 0111 \rightarrow 7 \\
 + 0011 \rightarrow 3 \\
 \hline
 1010 \rightarrow 10
 \end{array}$$

Diagram 5.2. Binary addition.

This works the same in binary because if we have:

$1 + 1$ gives 10, which is binary for 2.

So the sum bit for that column is 0, and the carry is 1.

$1 + 1 + 1$ gives 11, which is binary for 3. So the sum bit is 1, and the carry is 1.

How do we build a circuit using logic gates that performs this standard addition algorithm?

Well, let's start with the rightmost column. If we think about it, all the possible states are:

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

So just $0 + 0$, $1 + 0$, $1 + 1$, or $0 + 1$. That's it! If we can make a tiny circuit that takes two inputs, and produces two outputs that match these combinations, we have added the first column.

This is called a half adder. A half adder adds two bits, but it does not handle a carry-in value. That is the job of a full adder.

Let's first build this half adder.

Let's start by computing the sum, not the carry-out.

This is what we want our circuit to do:

A	B	Sum
0	0	0
0	1	1
1	0	1
1	1	0

The sum is 1 only when exactly one input is 1.

This is called XOR short for exclusive OR.

If we combine an OR gate and a NAND gate, and AND them together we get XOR:

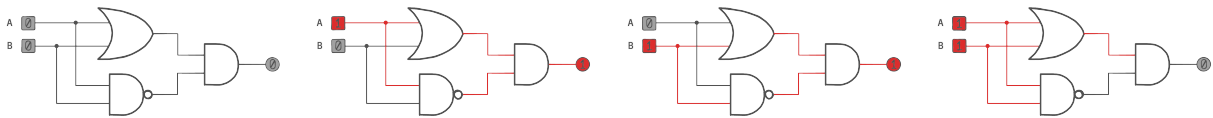


Diagram 5.3. Half adder sum / XOR.

OR checks that at least one input is on, and NAND makes sure that both inputs are not on.

Here is how an XOR gate looks:

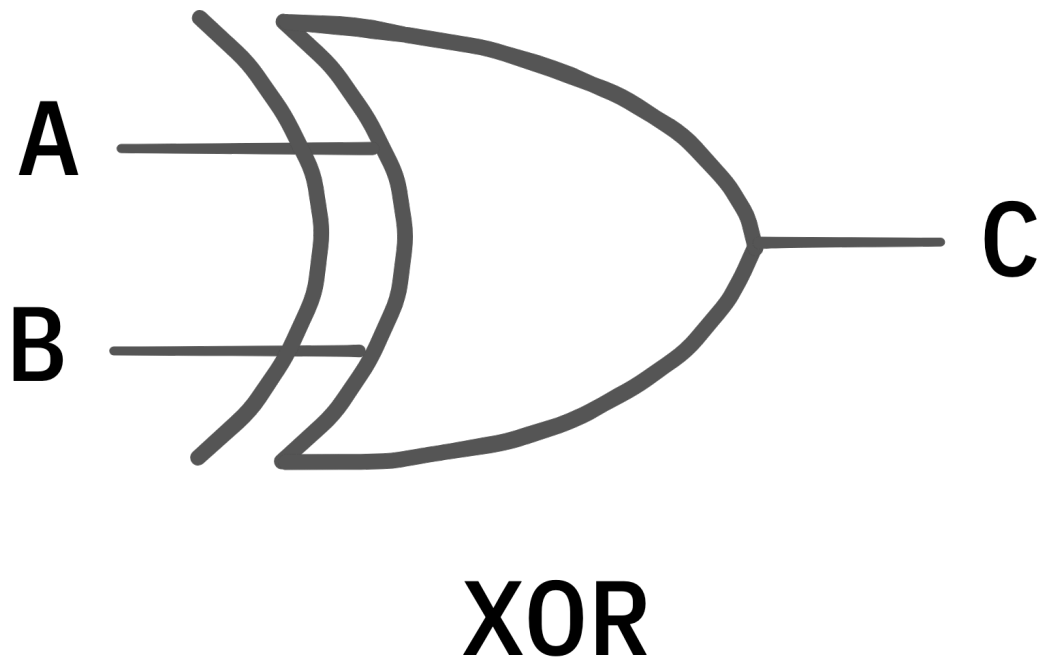


Diagram 5.4. An XOR gate.

Now let's do the carry value. The carry is simple! We only want to carry if we are doing $1 + 1$, so we just use an AND gate to check if both inputs are on.

Now here is our half adder:

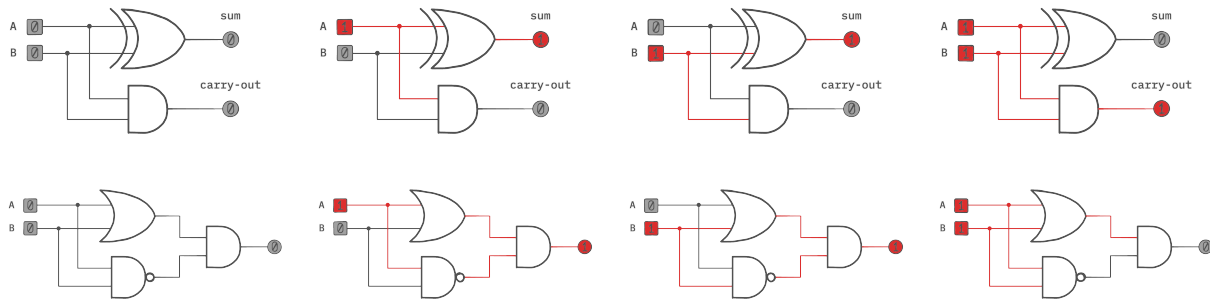


Diagram 5.5. A half adder.

As you can see it works! $0 + 0 = 0$, $1 + 0 = 1$, $0 + 1 = 1$, and $1 + 1 = 10$.

Now let's package up our half adder into a little box:

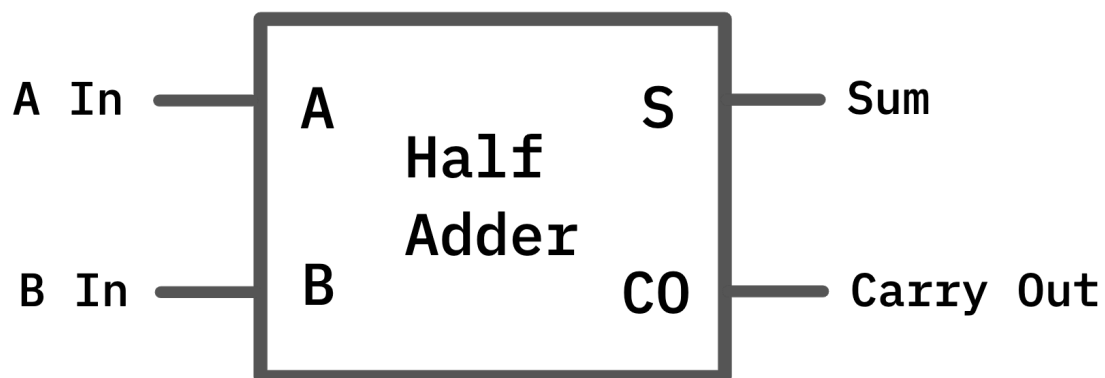


Diagram 5.6. A half adder chip.

Now that we have a half adder, we can add the rightmost column. That works because the rightmost column has no carry-in from a previous column. It only needs to add two bits.

So if we have a number like this:

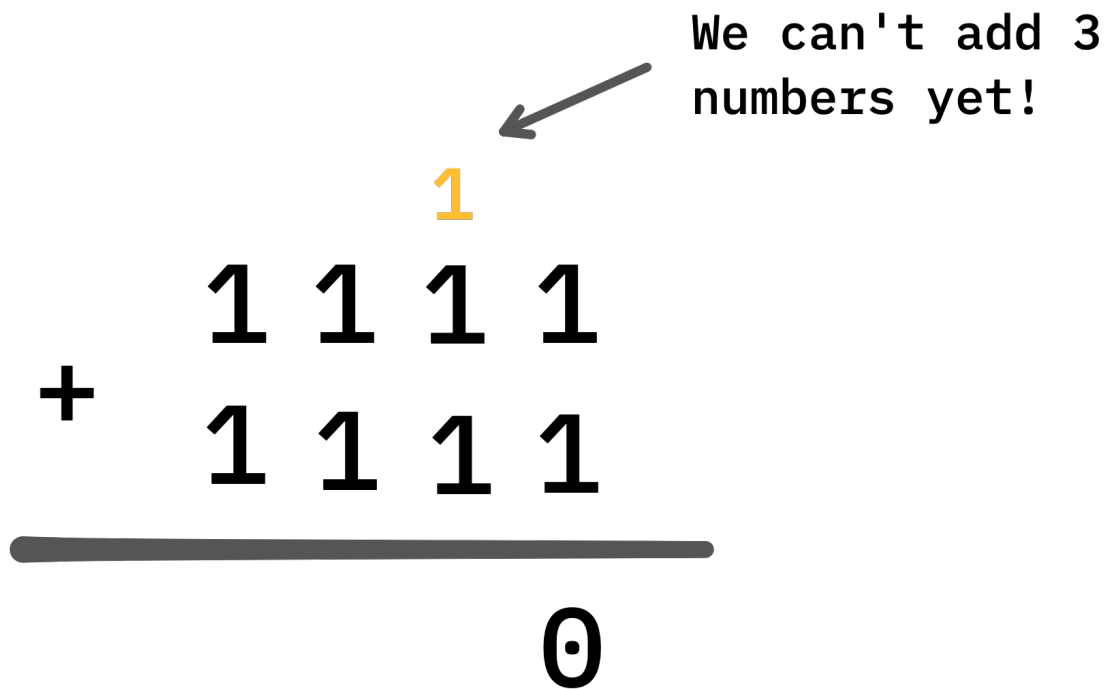


Diagram 5.7. The next column has to add two bits plus a carry-in.

The half adder can handle the first column: $1 + 1$. That gives us a sum bit of 0 and a carry-out of 1.

But now the next column has three things to add: $1 + 1 + 1$. The two original bits, plus the carry from the previous column.

A half adder cannot do that. It only accepts two inputs. To continue adding up the other columns, we need a circuit that can take in three inputs: A, B, and carry-in.

To add three binary numbers, we use two half adders and an OR gate:

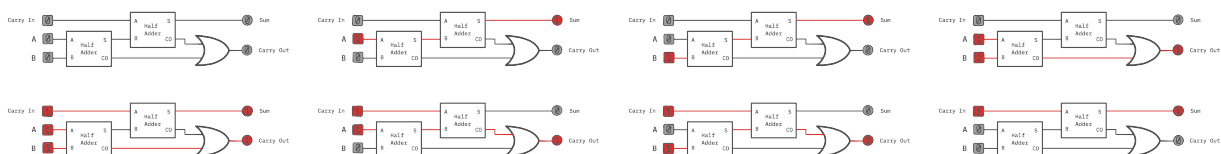


Diagram 5.8. A full adder.

This might look confusing at first. What if both half adders output a carry-out at the same time?

That actually never happens. If a half adder outputs a carry, the sum bit is always 0. So both are not able to output carries. Take a moment to think about this if you are confused.

So we can confidently OR the two carry outputs together. If either one is 1, the full adder's carry-out is 1.

Let's again package this up into a box:

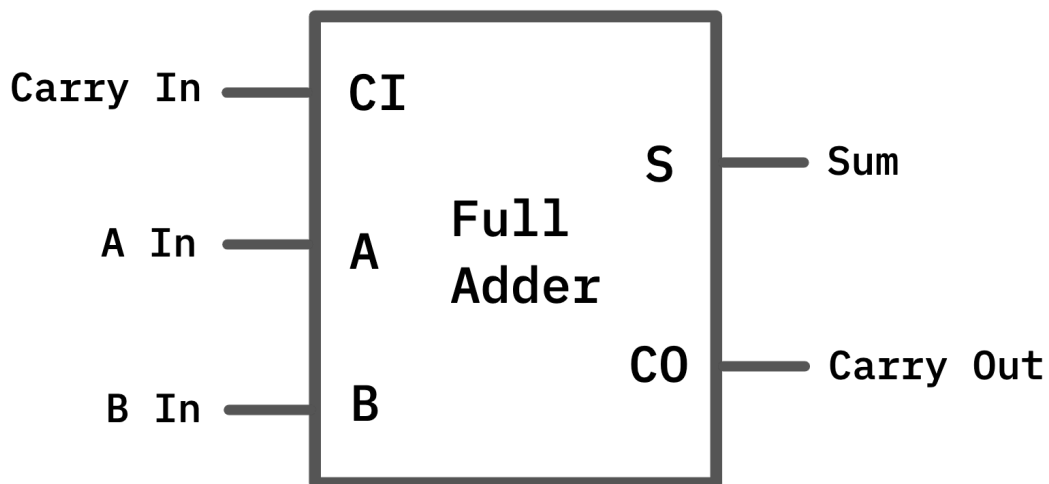


Diagram 5.9. A full adder chip.

We have made a full adder!

Now we can chain full adders together to add two 8-bit numbers. Since 8 bits make one byte, this is an adder that can add two one-byte numbers. One byte can represent any number from 0 to 255. That is 256 total combinations because 2^8 is 256.

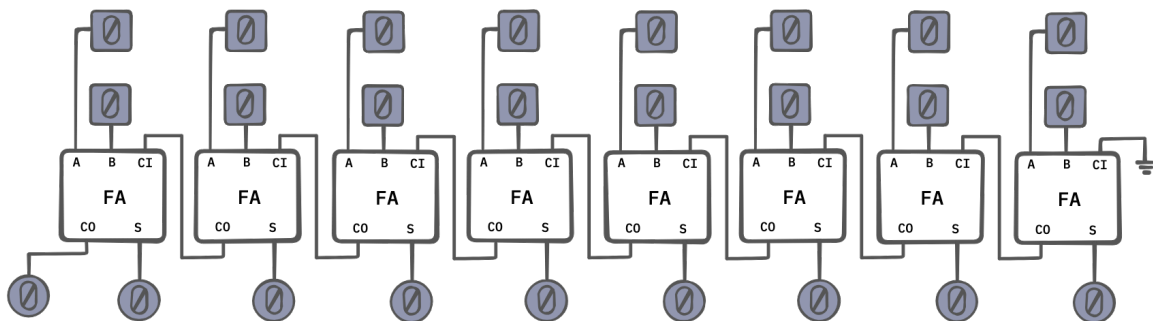


Diagram 5.10. An 8-bit adder.

Each full adder handles one column. The carry-out from one column becomes the carry-in for the next column. That is it! That is all addition is!

Keep in mind, carry-in for the first adder is set to ground, a.k.a. 0.

Also, notice how we have 9 outputs, not 8. That is because two 8-bit values can add up to a number greater than eight bits. It's like how adding two 2-digit numbers could result in a three-digit number for us. Like $50+50=100$.

Now let's package this up into a box once again:

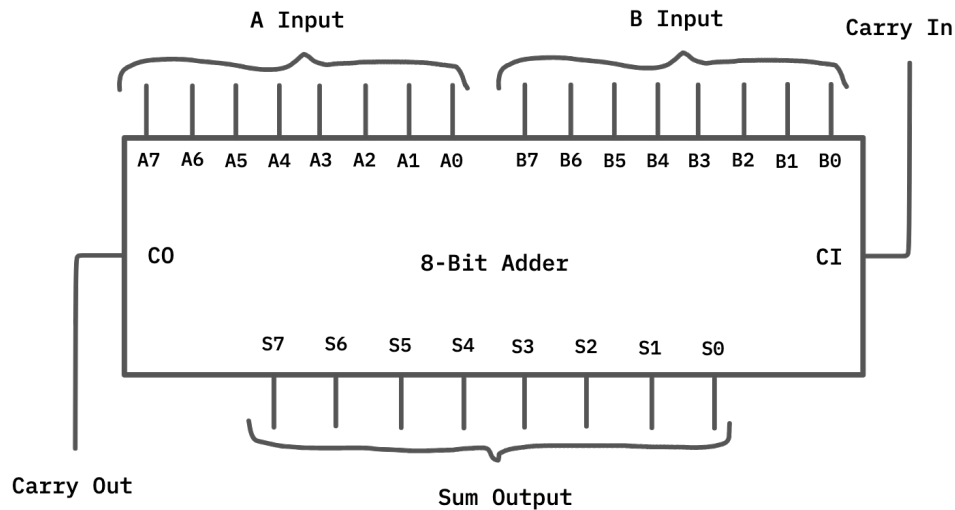


Diagram 5.11. An 8-bit adder chip.

Now we have the carry-out and carry-in as separate inputs and outputs and the whole adder nicely organized into this chip.

Let's have a look at some example problems:

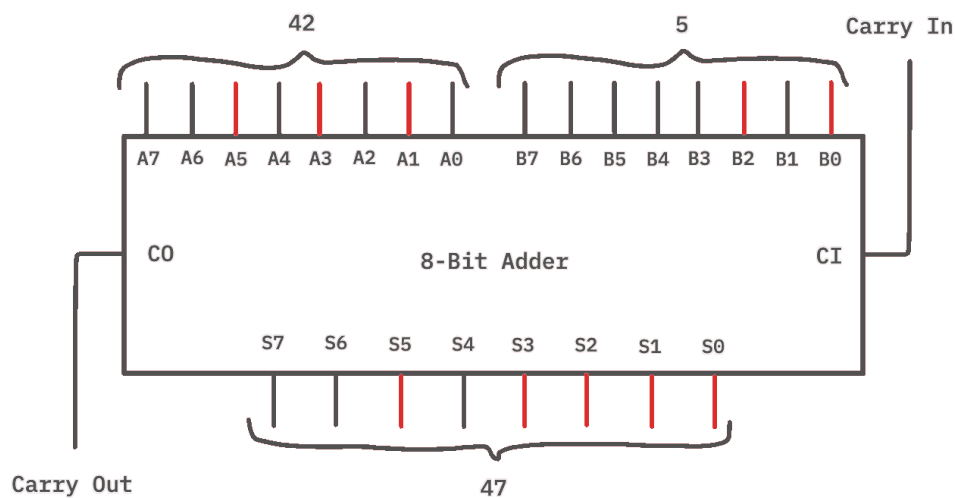


Diagram 5.12. Some examples on the adder.

As you can see in the third example, adding 1 to 255 turns every sum bit to 0 and turns the carry-out on.

This doesn't mean the adder got the wrong answer. In fact, $255 + 1$ is 1 00000000 in binary: eight 0 output bits, plus one extra carry-out bit on the left. If we only look at the one-byte output, the result looks like 00000000, or 0. If we also look at the carry-out, we can see that the real answer was 256.

That is called an overflow: the result was too large to fit inside one byte, so the extra information spilled out into the carry-out bit.

The adder can also produce little status wires, called flags.

For example, if the answer is 00000000, a ZERO flag can turn on. If addition spills past one byte, a CARRY flag can turn on. So 11111111 + 00000001 gives 00000000 with carry-out 1.

I don't want to go deep into flags yet. Just remember that the adder can output little yes/no facts about the sum. That matters later for instructions like "jump if zero." But let's not get ahead of ourselves.

We have just built addition! But we also need something else: storage.

For example, let's say we want to build a circuit that counts by ones, like 1, 2, 3, 4,...

The obvious idea is to feed the output of the adder back into one of its inputs. Start with 00000000, add 00000001, get 00000001. Feed that back in, add 00000001 again, get 00000010. Then 00000011, then 00000100, and so on.

That seems correct at first glance.

But there is a big problem. An adder just looks at its current inputs and computes an output.

So if we wire the output straight back into the input, there is no stable value anymore. The adder is basically being asked to make a number equal to itself plus one:

$\text{input} = \text{input} + 1$

That can never settle. As soon as the output changes, the input changes too, which means the output has to change again, which means the input changes again.

With relays, you might physically see this mess play out. With transistors, it would happen almost instantly.

There is no boundary between the old value and the new value.

There is no clean "step 1, step 2, step 3." No controlled flow or process.

So this is not enough. We need a circuit that can hold a value still, then update it only when we tell it to.

That is the next problem: memory.

6 Storing a Bit

To store a bit, we need to understand feedback. Feedback is simply feeding the output of a circuit into the input. There are two main kinds of feedback, unstable and stable. We just witnessed an example of unstable feedback, where feeding the output of the adder into its input resulted in messy and unpredictable behavior.

The other type of feedback is known as stable, because it can produce two stable states. Stable feedback is used to create circuits whose outputs aren't purely based on their inputs, but also based on what happened before. Stable feedback is exactly what we need to create memory.

The circuit that does this is called an SR latch. SR stands for set-reset. The value Q is the output we really care about. If it is 1, that means the latch is storing a 1; if it is 0, the latch is storing a 0.

The diagram also shows a second output written as a Q with a bar over it. That is just how engineers write $\text{NOT } Q$, pronounced "not Q ". It always holds the opposite of Q . I will write it as $\text{NOT } Q$ in the text.

The two inputs are SET and RESET, drawn as little buttons in the diagram: gray means not pressed, red means pressed. Pressing SET forces Q to 1 and pressing RESET forces Q to 0.

For this circuit to be used properly, set and reset should never be on at the same time.

The cool part is, if both set and reset are 0, then Q is whatever we last did to it! The output loops back into the circuit, so the current state keeps reinforcing itself. This is the basic concept behind memory.

This diagram should help this make sense:

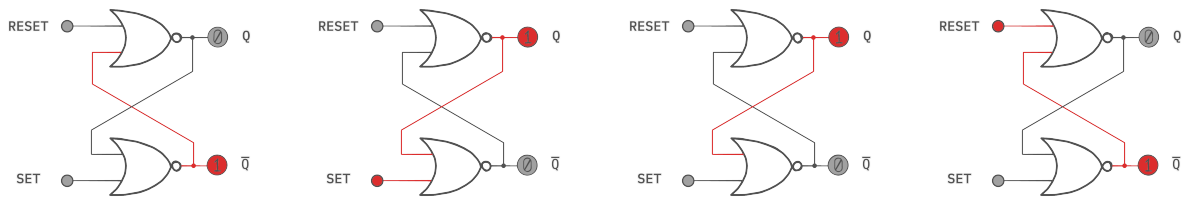


Diagram 6.1. An SR latch.

A simple way to think about this is:

If SET is on, the bottom NOR gate has to output 0, because one of its inputs is on. That makes NOT Q equal to 0.

Now the top NOR gate sees two 0 inputs: RESET is 0, and NOT Q is 0. So the top NOR gate outputs 1, making Q equal to 1.

Then even if we turn SET back off, the latch stays in that state. Q is still 1, which keeps forcing NOT Q to 0, and NOT Q being 0 allows Q to stay 1.

RESET works the other way. If RESET is on, it forces Q to 0, which allows NOT Q to become 1. Then even after RESET turns off, NOT Q keeps forcing Q to stay 0.

The circuit has state. Its output depends not only on the current input, but on what happened before.

Now that we have the core mechanism, let's refine the interface. All I mean by that is, right now SET and RESET are super clunky. While they demonstrate the mechanism, what we would really like to have is two inputs.

- Data (D)
- Enable (E)

When the enable wire turns on, Data gets stored in Q. Or in other words, when we turn the Enable wire on, Q mirrors D. Then when we turn E off, Q stays stable with whatever D was last.

This is much easier than fiddling with SET and RESET.

This type of latch is called a D latch, D meaning data. It can be made using the SR latch and a few extra logic gates.

It basically checks: if data is true and enable is true, set is true, and if data is false and enable is true, reset is true. That's it, so let's not worry about the exact implementation.

If you really want to know how it works, have a look at this site.

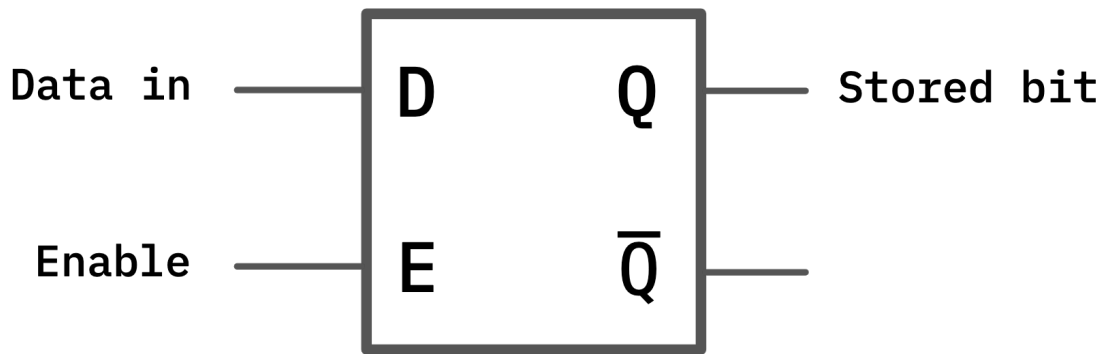


Diagram 6.2. D latch.

But we have a problem. Let's say we now try to use 8 of these D latches to hold the result from our adder, which would then feed back into the input for our accumulator. It still wouldn't work.

This is because let's say we have the enable wire hooked up to a button. When that button is pressed down, the enable wire is on, so $Q=D$ for that time. But if Q feeds back into the adder, and the result of the adder D changes quickly enough, Q can change again, jumping unpredictably based on how long we hold that button for.

If we want the accumulator to work correctly, we need the enable wire to turn on for an instant and then turn back off. That is just hard to do.

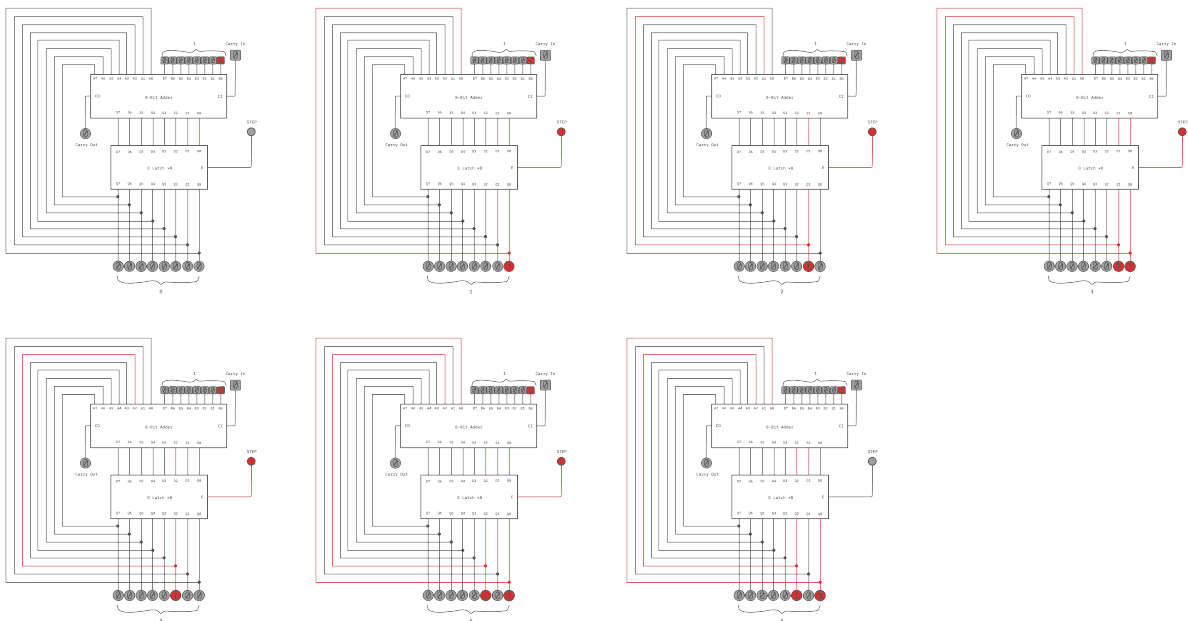


Diagram 6.3. D latch accumulator.

As you can see in this diagram, even pressing the button quickly jumps the result up by 5. With real transistors, even if you try to physically tap the button, it could count up by millions, overflowing these 8 bits thousands of times.

How long you hold the button decides the answer. It doesn't count in ones.

But what if we had a storage circuit that only copied D into Q at the exact instant E turns on?

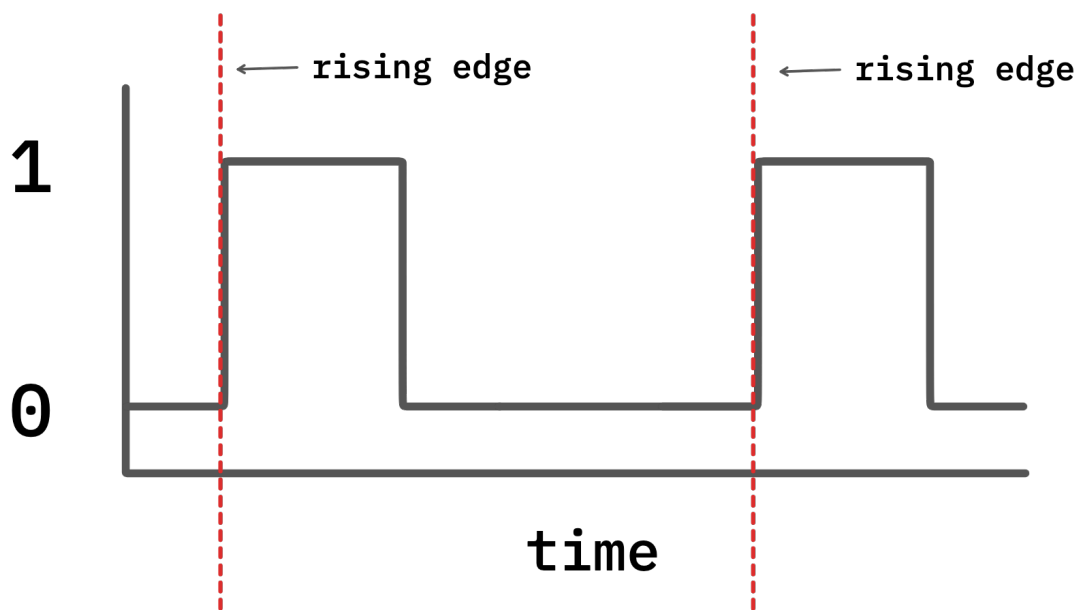


Diagram 6.4. The rising edge of a signal.

This graph shows the state of a wire. When the line is at the top, it is on. When it is at the bottom, it is off.

When a switch is flicked or a button is pressed, a transition happens. That transition is called an "edge", and when the wire turns from off to on, it is a rising edge.

Now what if we only set Q to D on that transition, at the rising edge? The edge is an instant of time, not a duration.

The circuit that does this is called a D-type edge-triggered flip-flop. This might sound like a mouthful, but D-type just means it takes in a data input, edge-triggered means it triggers on the edge of a signal, and flip-flop means it is a storage circuit similar to a latch, but usually edge-triggered.

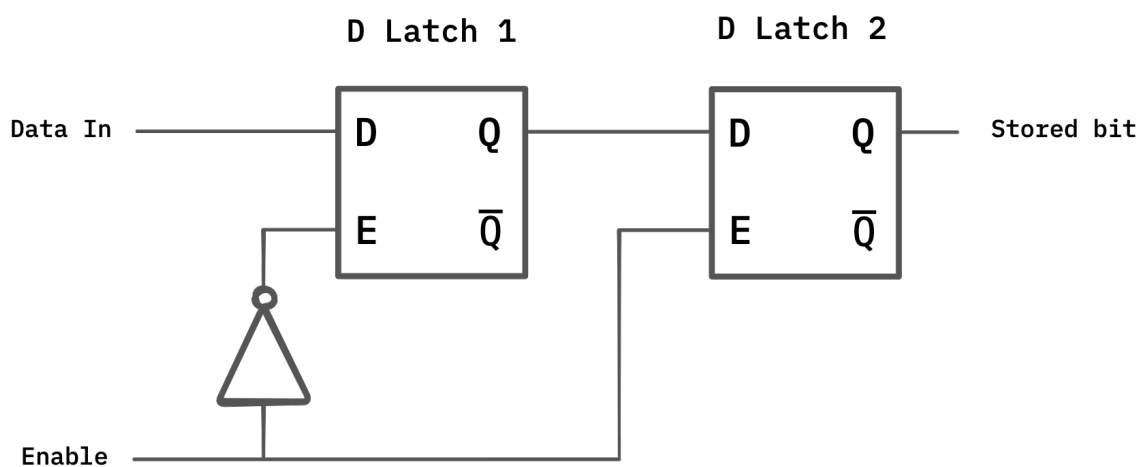


Diagram 6.5. A flip-flop.

How it works is, when the enable wire is off, the first latch mirrors D. That is because the NOT gate flips the enable signal, so the first latch sees it as on.

Then when enable turns on, the second latch stores the output of the first one. And because enable is now on, the first latch is locked, so it can't change!

So if D changes while enable is off, we are all good because the second latch is locked. But if D changes while enable is on, we are fine because the first latch is locked.

That is how this flip-flop works.

Here is one storage cell, which is just the flip-flop we showed above:

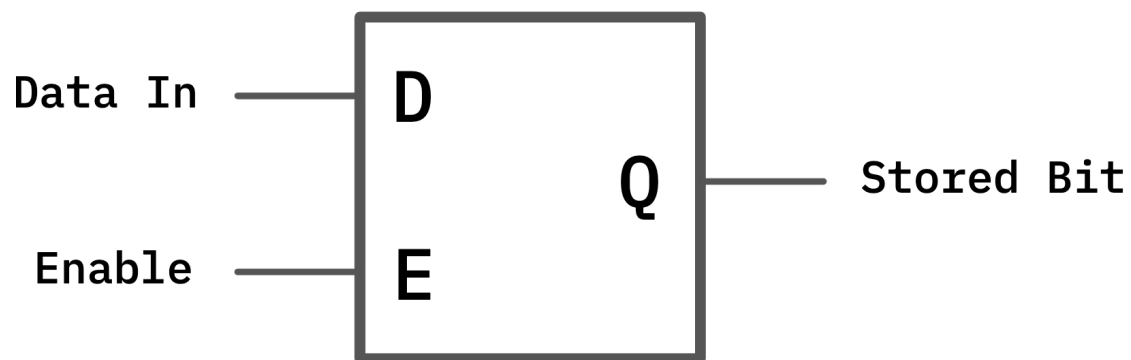


Diagram 6.6. A one-bit storage cell.

If we connect 8 of them side by side, we get one byte of storage:

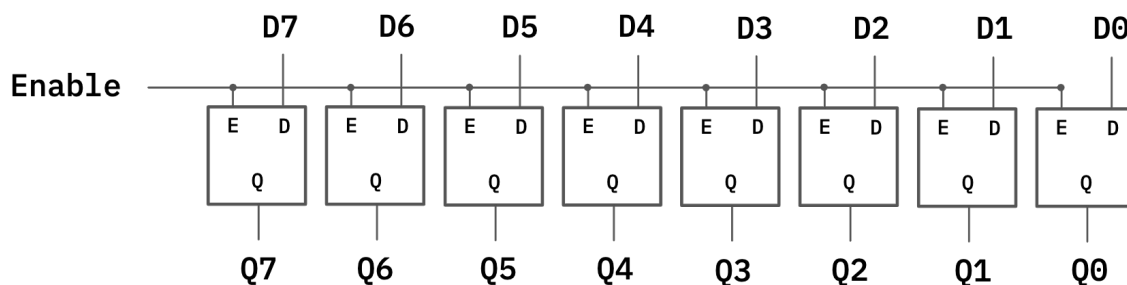


Diagram 6.7. Eight storage cells.

And we can put all that into a box called an 8-bit register:

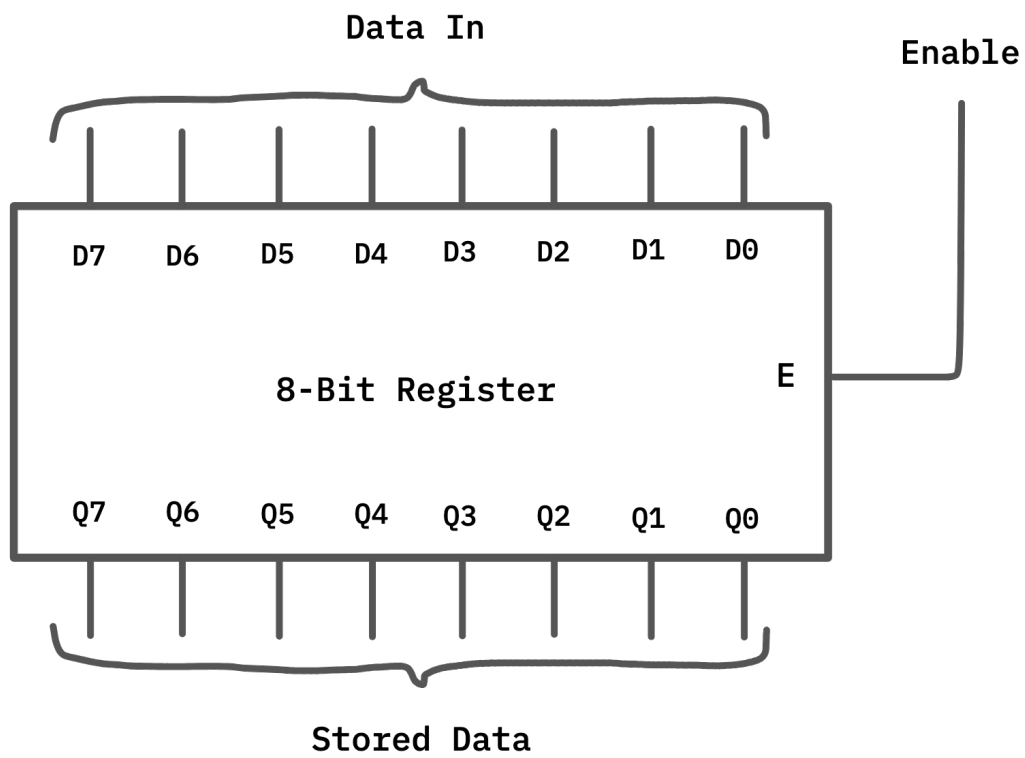


Diagram 6.8. An 8-bit register.

Now with this register, let's build a basic accumulator/adder circuit.

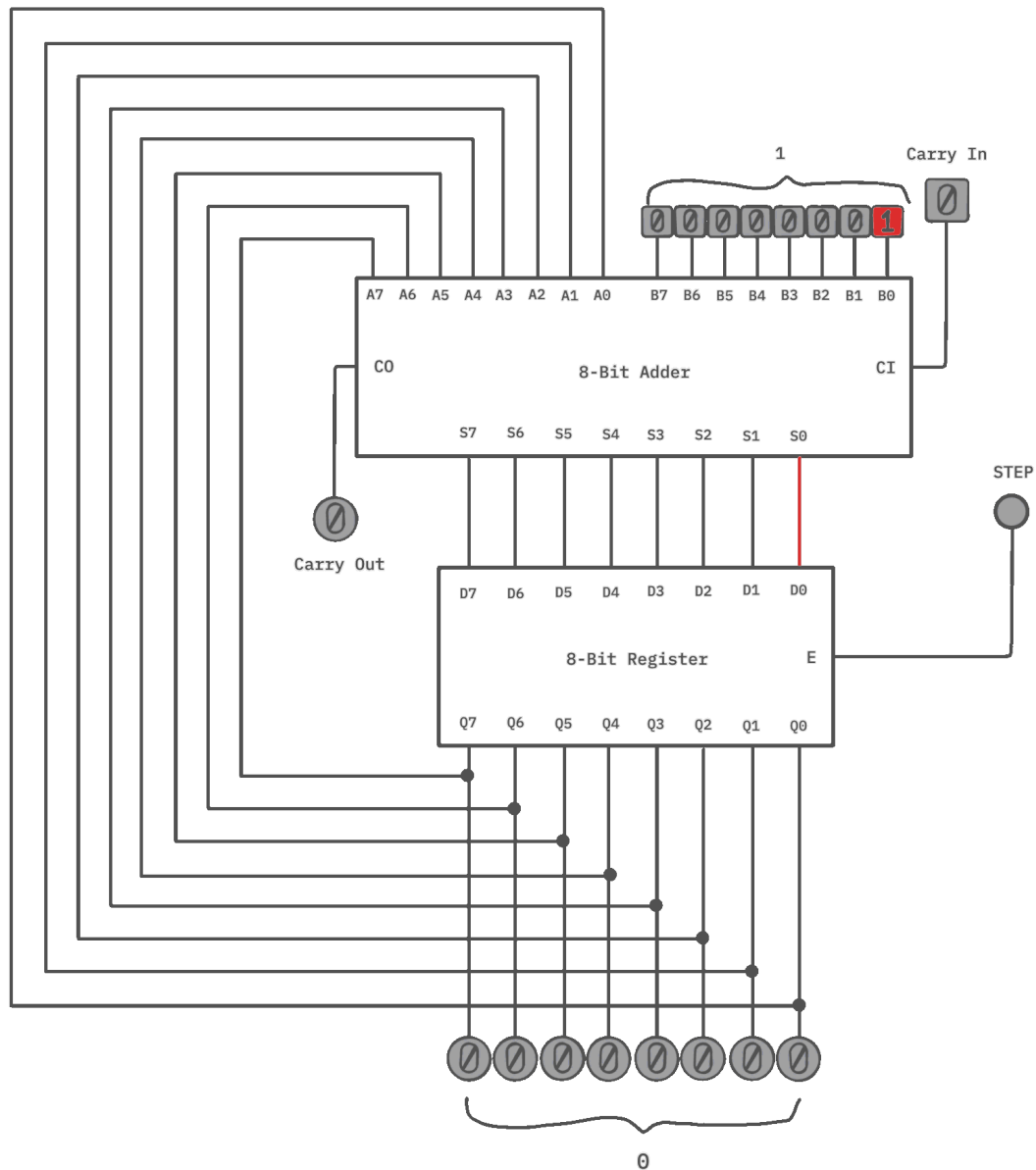


Diagram 6.9. Our full accumulator.

As you can see, the circuit kindly waits for us, and is incrementing by ones!

How this works is, when the STEP button is pressed, the output from the adder gets saved into the register on the rising edge of that press. This then changes the input to the adder, which changes its output, but the register holds its value because it only captures on the edge of the press. Holding STEP down does nothing special. So each press increments the register's value by, in this case, 1.

Now, a real computer would need to do these kinds of things millions and billions of times per second, and we don't have some human clicking a step button. What we have is a circuit that automatically goes on, off, on, off billions of times per second. This is called a clock. Each rising edge of the clock acts like one press of STEP.

Here is the basic concept of a clock:

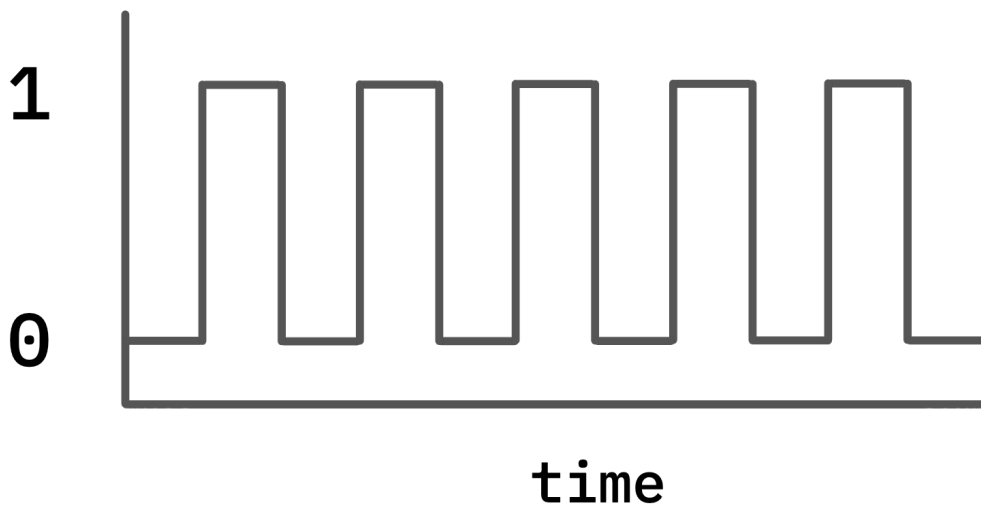


Diagram 6.10. A clock signal.

This repeating on-off behavior can be achieved in different ways. A rough toy example is feeding the output of a NOT gate back into its input, so the signal keeps trying to flip back and forth between on and off.

Real clocks are built in more sophisticated and reliable ways, often using crystals or other oscillator circuits. But we do not need to build the clock itself here. For now, we can treat it as a little box that repeatedly produces the same on-off signal.

Just imagine the new accumulator with a clock signal instead of a STEP button. I am too lazy to draw it for you.

We now have some storage. A register that can hold a byte, and update exactly when we want.

But registers on their own are not enough. We need to be able to move numbers between registers, the adders, and the main memory, which we will build later.

7 Buses

One simple solution to move bytes around would be to give every component its own bundle of 8 wires to every other component, but that would become a mess very quickly.

A simpler solution is to have one single 8-bit data highway, where components can put data on and take data off. This collection of 8 wires is called a bus.

One more thing, moving forward when I want to draw a collection of 8 wires, instead of drawing each wire, I will just draw a thick arrow that represents 8 wires.

To show the state of the wires, I can write a number in the arrow; the number 0 for example means the wires are all off, and the number 2 would mean the wires are 00000010 which is 2 in binary.

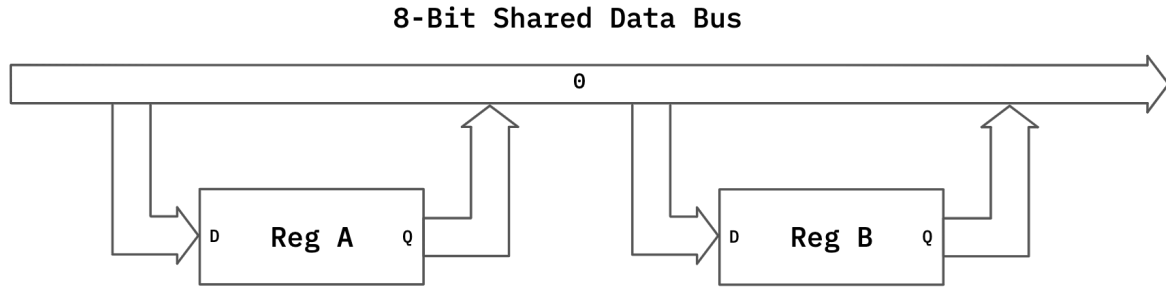


Diagram 7.1. Two registers sharing a bus.

But we have an issue: this diagram is technically not possible yet. If register A is outputting a value like 00000000, and it is connected to the bus, and then register B is outputting a value like 00000001, then the last wire will clash and short-circuit.

We need a way to connect these registers to the bus, but also let them get out of the way when they are not supposed to actively drive a wire to - or +, like we discussed previously.

We need a way to make wires "free" when we don't want to output anything.

Just setting the output wires to 00000000 is not enough. The point I am trying to make is that on a shared bus 00000000 is not nothing. It is actively driving the bus to -.

One clean way to solve this problem is by using something called a tri-state buffer. It has two inputs, E and D, which stand for enable and data.

We have seen E before, in the context of "enable writing" but now we are using E in the context of "enable outputting".

If E is on, the output will just be whatever D is, so either 0 or 1. If E is off, no matter the value of D, the output will be Z.

Z means the buffer's output is disconnected from the bus. It is not driving the bus to + or -, so another component can safely drive the bus without a clash or short-circuit.

Or in other words, this buffer is not touching the wire. 0 is very different: the component is actively pulling the wire down. So we have three states:

- 1: driven high
- 0: driven low
- Z: disconnected

This relay diagram of how a tri-state buffer works should make this concept crystal clear.

Also I have drawn everything the output wire is currently touching in yellow. Yellow is just there so you can follow the path with your eyes, it doesn't mean anything.

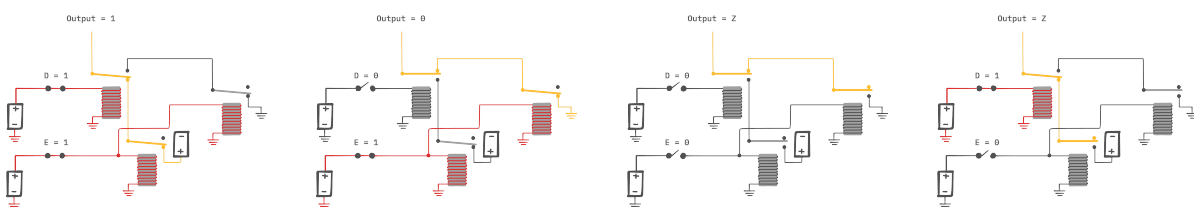


Diagram 7.2. A tri-state buffer built with relays.

This looks complicated, so let me break it down.

First, ignore the two relays on the right and look only at the D relay at the top. Its arm is attached to the output wire, and it works just like the output driver from before. When D is 1, the arm is pulled down onto the wire that leads toward the battery, +. When D is 0, the arm goes up onto the wire that leads toward ground. Remember, ground is just the - side.

The important idea is that neither of those wires is directly connected to + or -. Each one has a relay between it. Both of those relays are controlled by E.

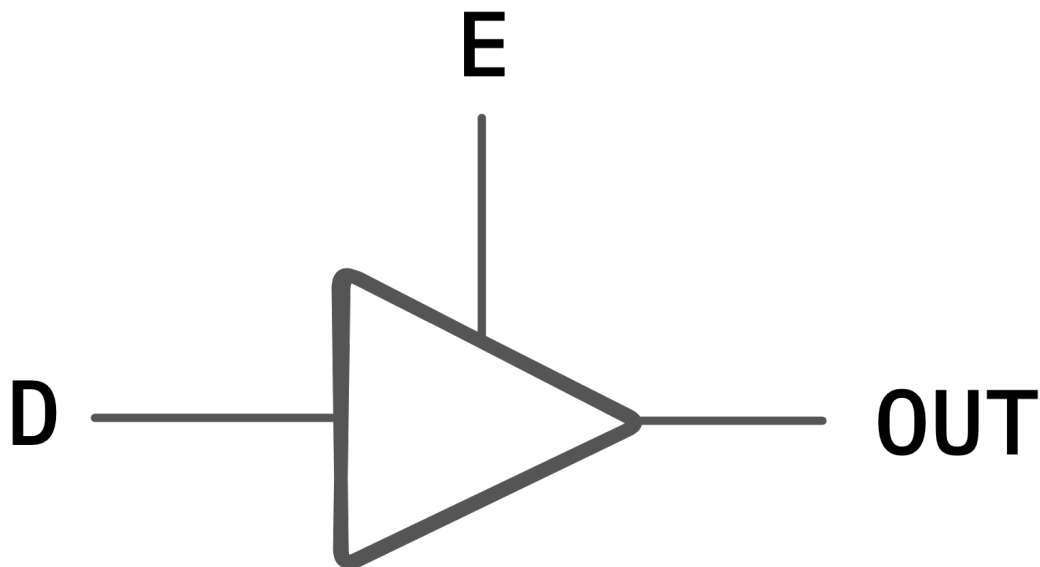
When E is 1, both gaps close. The output is now connected to whichever side D picked, so it is driven to 1 or 0.

When E is 0, both relays touch a point connected to nothing. Both wires lead to a dead end. The output wire is touching nothing. That is Z.

So we have three states:

E	D	Output
1	1	1
1	0	0
0	0	Z
0	1	Z

This is the logic gate diagram for a tri-state buffer:



Tri-State Buffer

Diagram 7.3. A tri-state buffer logic gate.

Now let's address this enable conundrum. We now have two uses for the word enable, with completely different meanings and contexts. One means enabling writing, and the other means enabling output. From now on, we will use two separate terms to avoid confusion: **WRITE** and **OUT**.

So we can make a new type of register, one with **WRITE**, **OUT**, data, and **Q**, the stored bits 0-7. By data, I simply mean the input data that we can store when **WRITE** is enabled.

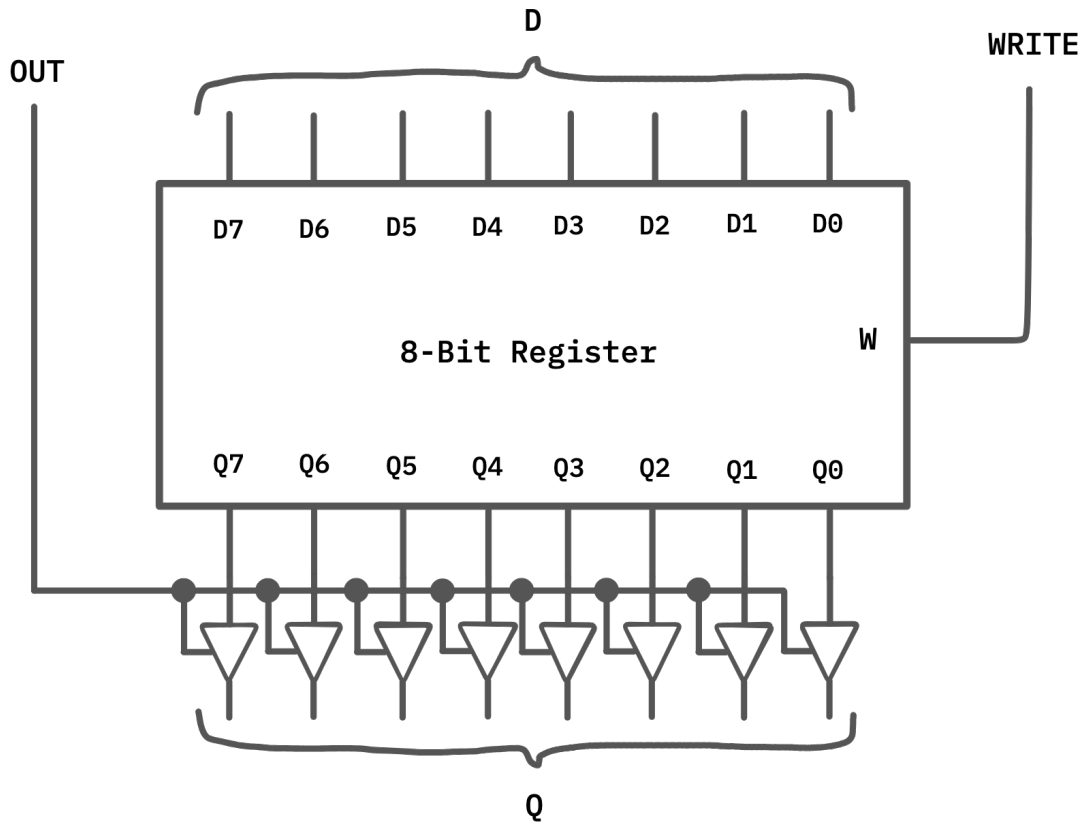


Diagram 7.4. Our register with an **OUT** input.

We are just connecting **OUT** to all of the enables in the tri-state buffers. So if **OUT** is 0 **Q** will be all Z and if **OUT** is 1, **Q** will be whatever is stored in the register.

With that, we can use these new registers with a common bus to move data.

Here is an example where the content of register A gets copied into register B.

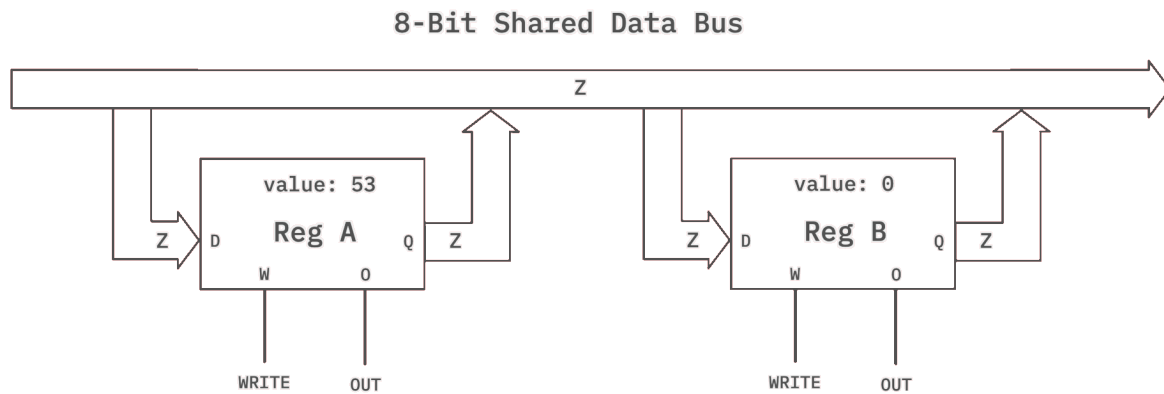


Diagram 7.5. Copying register A into register B through the shared bus.

I have some text inside the register that shows what it is storing. We of course have W and O which are WRITE and OUT as well as D and Q which are the inputs and outputs.

Of course, on the second frame, when OUT of register A is enabled the D wires of both registers are also going to be 53 because they are directly connected to the bus.

By the end of this sequence, we have copied the value 53 to register B! We can have many more registers sharing a common bus, as long as only one is driving the bus at a time.

The next problem is organization and scale. How do we organize many stored bytes so the machine can choose one slot, read it, and write back to it?

8 Organizing Data (REDO, because I added BUSES)

We want to build a system that organizes data into the structure of Otto's cabinet slots.

This system is known technically as RAM: Random Access Memory. It is called RAM because when the CPU wants to access a slot, it just knows the number and can access any slot at will. It is not like flipping through a book looking for the right page. It is more like grabbing a book from a bookshelf, where you already know the book.

Now let's think about exactly what we would want this RAM chip to do.

- address: the slot we wish to access
- WRITE: whether we want to write a value to this address
- data in: the value we would like to write
- data out: the value we would like to read

Okay, let's make this more precise. We are going to build a minuscule 16-byte RAM: 16 addresses, with each address storing one byte. This design can be scaled up easily.

Our address will be 4 bits long, because 2^4 is 16. Just enough to represent every single address.

Now we ideally don't want a tall stack of 16 registers, we want a nice grid pattern.

Thus, we will use 2 out of the 4 bits for the row, and the other 2 bits for the column.

2 bits can store 4 values, so we will have a 4×4 array of memory, which is 16 total values!

When we select an address, we want RAM to automatically put that register's stored byte onto data out. If WRITE is on, then on the edge of WRITE turning on, that register stores data in.

Let's start with building a simple decoder. This decoder will take 2 bits of our address and, based on that number, turn on exactly one out of 4 wires.

In the diagram, the two input bits are labeled A1 and A0. A1 is the bigger bit, the 2's place. A0 is the smaller bit, the 1's place.

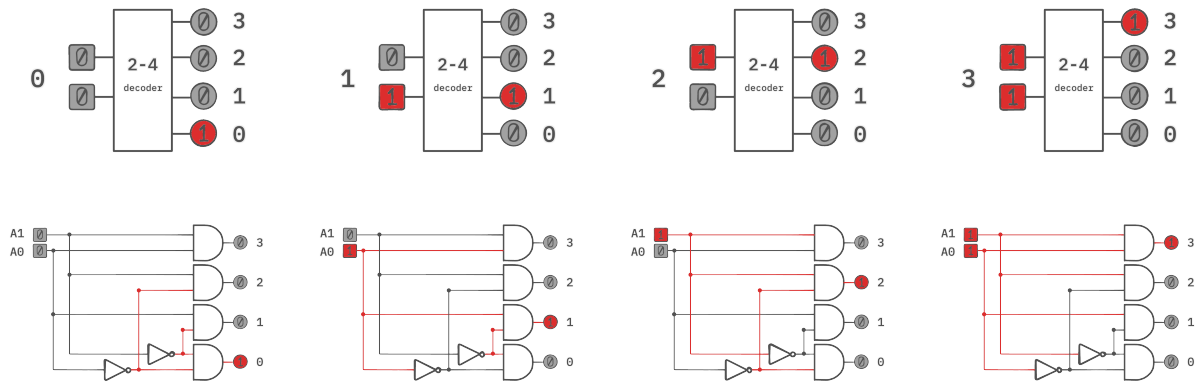


Diagram 7.1. How a decoder works.

As you can tell, no matter the inputs, exactly one output wire is on at a time.

We use one 2-to-4 decoder for the rows and another 2-to-4 decoder for the columns. Where the selected row and selected column cross, that is the byte we want to target.

This diagram shows a few addresses as examples. Each address gets its own little intersection. Each address from 1-16 has its own spot.

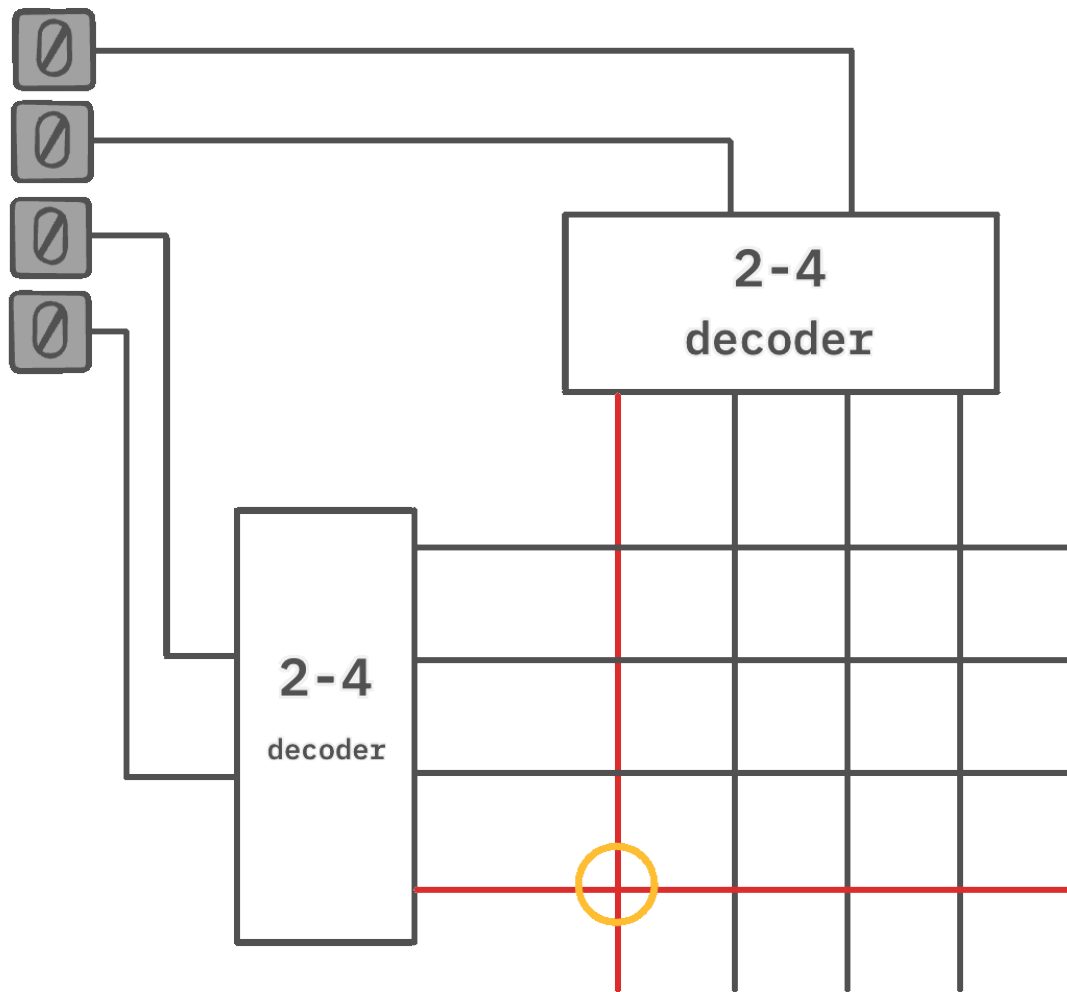


Diagram 7.2. Where the row and column meet.

How a decoder works is extremely simple. It just uses a bunch of logic gates to ask these simple questions.

- If 00 -> turn on wire 1
- If 01 -> turn on wire 2
- If 10 -> turn on wire 3
- If 11 -> turn on wire 4

Here is how it works if you care:

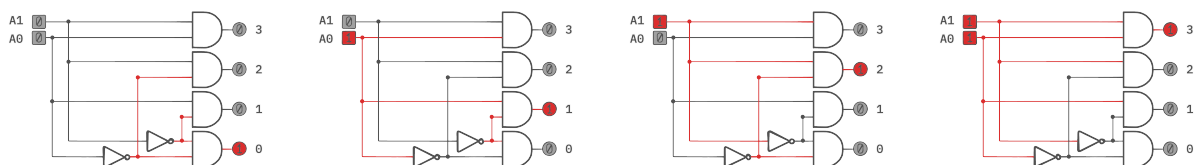


Diagram 7.3. 2-4 decoder internals.

One more thing, moving forward when I want to draw a collection of 8 wires, instead of drawing each wire, I will just draw a thick arrow that represents 8 wires. So instead of our previous register diagram, we would have something like this:

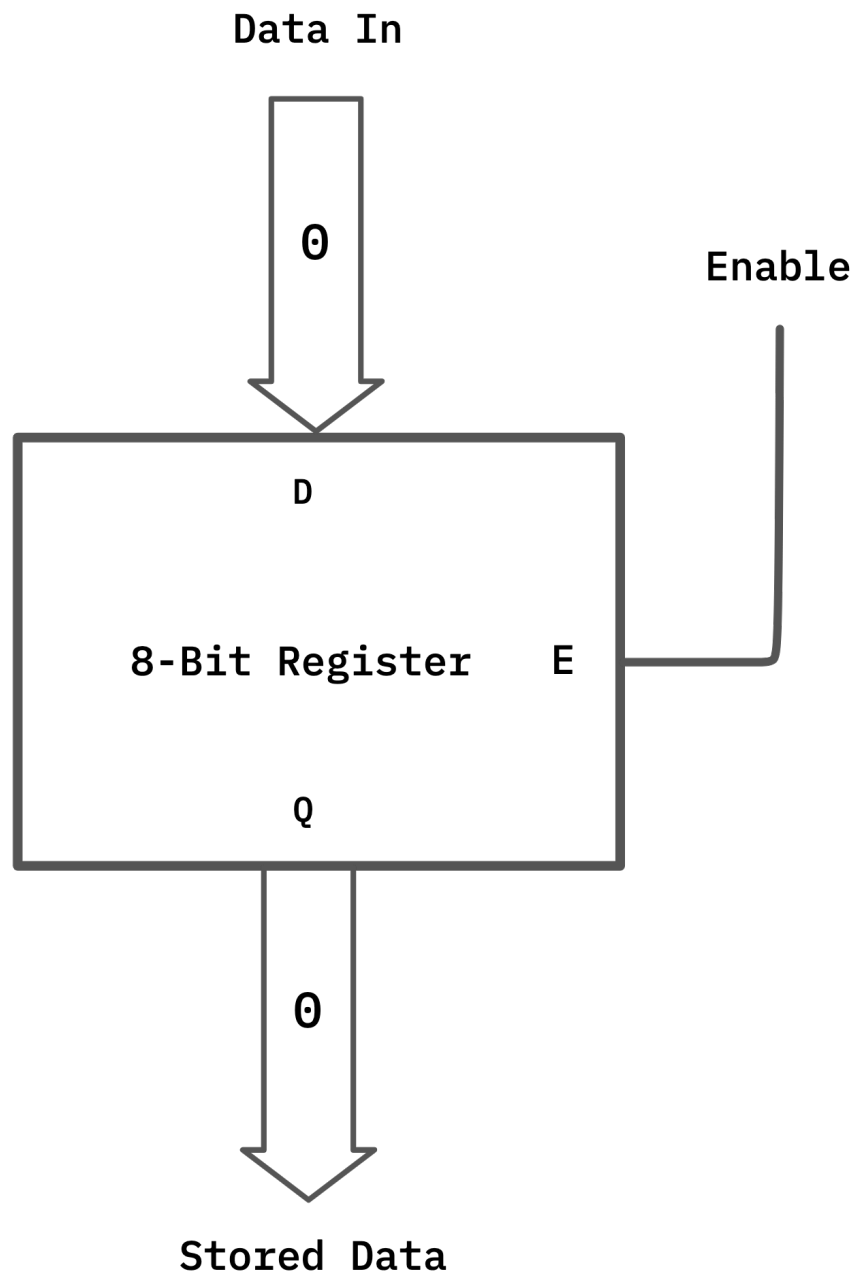


Diagram 7.4. An 8-bit bus.

To show the state of the wires, I can just write a number in the arrow; in this case, the number 0 means the wires are all off.

Okay, two more things we need to cover before I can show you the RAM diagram. First, let's add one more input to our register:

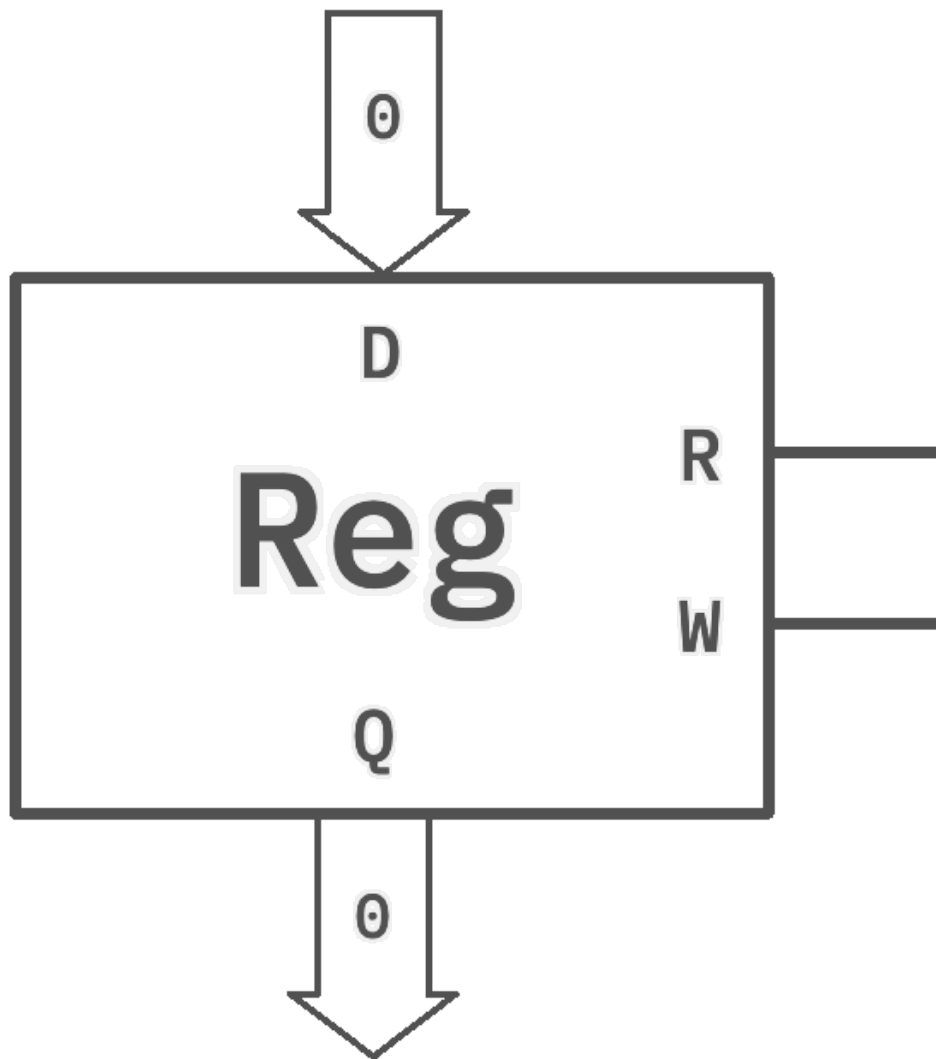


Diagram 7.5. A register with READ control.

These are our simple register diagrams that will be used in the RAM diagram later. R is READ and W is WRITE. It is, of course, an 8-bit register.

So now the slot has two control inputs: WRITE and READ. We are already familiar with WRITE which works like the previous enable wire, and now READ controls whether the slot can output its stored value.

The register's stored byte is sitting on eight output wires, Q0 through Q7. Before that byte leaves the slot, each bit is ANDed with READ.

Let's say $Q = 01011011$. If READ is 0, every bit gets ANDed with 0, so the slot outputs 00000000. But if READ is 1, every bit passes through unchanged, so the slot outputs 01011011.

Second thing. In our RAM design only one register will be selected at a time, and we need to combine all the outputs onto one bus that will show the output. To do this, we can just OR the values of each gate when we need to combine.

This works because all the gates but one will be 0.

```
register 1: 00000000
register 2: 00000000
register 3: 01010111
register 4: 00000000
output:    01010111
```

So if we OR all of these buses together we just get the value of the enabled bus.

In the later diagrams, when two buses merge through a blue connector, that means their bits are ORed together; they are not literally connected.

Honestly? That's it. We can use two decoders, sixteen registers, some output wires and some input wires all mashed together with some extra logic gates and BOOM! We have some RAM.

You might have noticed a few oddities in this diagram. First, I changed E to W, because here the enable input specifically means "write enable." The register should only copy data in when this slot is selected and WRITE is on.

I also draw an AND gate taking an 8-bit bus and one normal wire. That is just shorthand for eight small AND gates in parallel: Q0 AND selected, Q1 AND selected, Q2 AND selected, and so on. In other words, the selected slot is allowed to put its stored byte onto data out, while the other slots output 0.

Now we have built Otto's abacus, desk drawers and upstairs cabinet, all working and functional!